SHARP LOCAL WELL-POSEDNESS FOR THE NOVIKOV-VESELOV EQUATION

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Abstract. We establish local well-posedness of the Novikov–Veselov equation, at every fixed value of the energy parameter, for initial data in $H^s(\mathbb{R}^2)$ in the entire subcritical Sobolev range $s > -1$. The scaling threshold is therefore reached, and this is, as far as we know, the first time the Sobolev theory of a quadratic two-dimensional dispersive equation has been brought down to its critical exponent. The argument is perturbative and does not invoke the integrable structure: an infinite normal-form expansion indexed by binary trees is combined with a gauge transform that absorbs the boundary terms behind the previous obstructions, and the resulting gauged equation – carrying nonlinear contributions of every order – is closed by a contraction in a Fourier restriction space. The analytic engine is a family of frequency-restricted multilinear bounds, supported by a detailed description of the two-dimensional resonance geometry attached to (NV).

1. Introduction

1.1. **Setting of the problem and main results.** We consider the Novikov–Veselov equation on $\mathbb{R}^2$,

$$\partial_t u + (\partial^3 + \overline{\partial}^3)u + \frac{3}{4}(\partial(u\overline{\partial}^{-1}\partial u) + \overline{\partial}(u\partial^{-1}\overline{\partial}u)) + E(\overline{\partial}^{-1}\partial^2 u + \partial^{-1}\overline{\partial}^2 u) = 0, \tag{NV}$$

with initial data $u_0 \in H^s(\mathbb{R}^2)$. Here $\partial = \frac{1}{2}(\partial_x + i\partial_y)$ and $\overline{\partial} = \frac{1}{2}(\partial_x - i\partial_y)$ are the Wirtinger derivatives, and $E \in \mathbb{R}$ is the *energy* parameter.

A natural starting point for the analysis of (NV) is its scaling. If u solves (NV) with energy E, then

$$u_\lambda(t, x) = \lambda^2 u(\lambda^3 t, \lambda x), \quad \lambda > 0, \tag{1.1}$$

is again a solution, now with energy $\lambda^2 E$. At $E = 0$ this is a genuine symmetry of the equation, and the homogeneous norm preserved by (1.1) is exactly $\dot{H}^{-1}(\mathbb{R}^2)$; thus $s_c = -1$ is the critical Sobolev exponent for the zero-energy problem. For $E \neq 0$ scaling is broken, but as far as low-regularity questions are concerned the same exponent $s_c = -1$ continues to mark the conjectural threshold: indeed, the work of Angelopoulos [3] excludes analytic data-to-solution maps below $s = -1$ regardless of the value of E. The aim of the present paper is to reach this threshold by perturbative means, that is, without appealing to the integrable structure of (NV).

Let us briefly recall the origin of the equation. (NV) appears, in implicit form, in the work of Novikov and Veselov [44] (and, simultaneously, of Manakov [32]) as a $(2 + 1)$-dimensional completely integrable counterpart of the Korteweg–de Vries equation, built from the two-dimensional inverse scattering problem for the Schrödinger operator at a fixed energy. The one-dimensional reduction obtained by removing the y-dependence recovers KdV exactly, which places (NV) on the same shelf as the Kadomtsev–Petviashvili and Zakharov–Kuznetsov hierarchies among higher-dimensional dispersive generalizations of KdV; we refer the reader to [9, 41, 19, 20, 18, 26] for representative results in those

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models. Compared with them, however, the nonlocal multipliers of (NV) and the rather involved geometry of its quadratic resonances give the equation a distinct analytic flavour. There is also a connection at the level of physical models: after anisotropic rescaling, the formal limits $E \to \pm\infty$ produce KP-I and KP-II respectively, see [23, 24].

The equation can be approached from two rather different angles. On the integrable side, inverse scattering, Miura-type transforms, and the modified Novikov–Veselov hierarchy yield existence and long-time results that exploit the algebraic structure of the equation in a decisive way. The dispersive–perturbative side, by contrast, treats (NV) as a semilinear equation and tries to push robust Fourier-analytic methods as low in regularity as they will go; whether this second viewpoint is in itself sufficient to recover the full subcritical Sobolev theory is the question we settle here, in the affirmative.

On the integrable side, solutions have been constructed through scattering techniques in [6, 43, 38, 36]. The zero-energy case is amenable to the Miura map of Bogdanov [5],

$$\mathbf{M}[u] = 2\partial u + |u|^2,$$

which carries solutions of the modified Novikov–Veselov equation

$$\partial_t u + (\partial^3 + \overline{\partial}^3)u + \frac{3}{4}\Big(\partial(u\overline{\partial}^{-1}\partial|u|^2) + \overline{\partial}(u\partial^{-1}\overline{\partial}|u|^2) + u\partial^{-1}\overline{\partial}(\overline{u}\overline{\partial}u) + u\overline{\partial}^{-1}\partial(\overline{u}\partial u)\Big) = 0, \tag{mNV}$$

subject to the constraint $\partial u = \overline{\partial} u$, to solutions of (NV) at $E = 0$. Since (mNV) is cubic, its low-regularity theory is in a much better shape: Schöttdorf [39] established small-data global well-posedness in the critical space $L^2(\mathbb{R}^2)$, in line with analogous critical results for other $2d$ cubic models [35, 15]. Nachman, Perry and Tataru [2] subsequently lifted Schöttdorf's theorem to large data. Through the Miura map this provides existence for the zero-energy version of (NV) for a specific class of large data in $H^{-1}(\mathbb{R}^2) + L^1(\mathbb{R}^2)$.

A parallel mechanism is well known for KdV [33, 21, 12, 22, 10, 25], but the analogy is only partial: for KdV, complete integrability provides coercive conservation laws, and these eventually yield a global theory down to $H^{-1}(\mathbb{R})$, sharp by [34]. For (NV) the conserved quantities fail to be coercive, in accordance with the existence of blow-up solutions [40]. The integrable picture is thus extremely useful, but does not on its own settle the semilinear low-regularity Cauchy problem.

On the contraction-argument side, the picture before the present work was the following. In the regime $E \neq 0$, Kazeykina and Muñoz [23, 24] obtained local well-posedness for $s > \frac{1}{2}$ via stationary-phase estimates tuned to the dispersive symbol, and Angelopoulos [3] attained the endpoint $s = \frac{1}{2}$ in Bourgain-type spaces. For $E = 0$, where the resonance relation simplifies, Adams and Grünrock [1] reached $s > -\frac{3}{4}$. None of these results comes close to the scaling threshold $s = -1$, and Angelopoulos showed in addition that the data-to-solution map cannot be analytic at the origin below $s = -1$, so fixed-point methods are bound to fail there. The subcritical gap left open by these works is therefore

$$-1 < s \leq \tfrac{1}{2} \quad \text{for } E \neq 0, \qquad -1 < s \leq -\tfrac{3}{4} \quad \text{for } E = 0.$$

Our main theorem fills this gap entirely.

Theorem 1.1. *Fix $E \in \mathbb{R}$ and $s > -1$. For every $u_0 \in H^s(\mathbb{R}^2)$ there exist a time $T = T(\|u_0\|_{H^s}) > 0$ and a unique function $u \in C([0,T]; H^s(\mathbb{R}^2))$ with the following property: for any sequence $u_{0,n} \in \mathcal{S}(\mathbb{R}^2)$ with $u_{0,n} \to u_0$ in $H^s(\mathbb{R}^2)$, the corresponding smooth solutions u_n to* (NV) *– provided by* [3, 23, 24] *– are defined on $[0,T]$ and satisfy $u_n \to u$ in $C([0,T]; H^s(\mathbb{R}^2))$. The resulting data-to-solution map $u_0 \in H^s(\mathbb{R}^2) \mapsto u \in C([0,T]; H^s(\mathbb{R}^2))$ is analytic.*

Remark 1.2. The conclusion of Theorem 1.1 is purely semilinear: integrability plays no role in its proof, and the statement is uniform in the energy $E \in \mathbb{R}$. To our knowledge this is the first instance of a quadratic two-dimensional dispersive model whose Sobolev well-posedness theory has been pushed to the scaling exponent by perturbative methods alone. A companion result in [11] reaches a comparable level for KdV, but in the Fourier–Lebesgue scale $\mathcal{F}L^{s,p}$ rather than in the strictly stronger Sobolev scale that we use here.

Remark 1.3. The endpoint $s = -1$ is left open, and the reason is structural rather than technical: it is rooted in the *quadratic* character of (NV). Critical well-posedness has been achieved for several *cubic* $2d$ dispersive models – (mNV) [39], KP-II [19], the Dysthe equation [35], and the modified Zakharov–Kuznetsov equation [15] – using the Koch–Tataru U^2–V^2 machinery [29]; but no analogous endpoint statement is available for a quadratic $2d$ dispersive equation. Here the obstruction is explicit: in the gauged expansion of (NV) every order $j \geq 1$ reaches frequency criticality *simultaneously* at $s = -1$, generating arbitrarily many logarithmic divergences. The simultaneity is absent in cubic $2d$ models, which is precisely why the endpoint becomes reachable there.

1.2. **Overview of the method.** The strategy is shared with our companion paper on KdV [11], but the outcome is qualitatively different: where the KdV analysis produces an improvement in the Fourier–Lebesgue scale, the same mechanism applied to (NV) reaches the scaling threshold of its native *Sobolev* theory. Three ingredients are used in concert: an infinite normal-form expansion, a gauge transform built from that expansion, and a multilinear theory of Fourier-restriction type. The normal-form component descends from the work of Babin–Ilyin–Titi [4] and its later refinements [17, 30, 37, 28]; the multilinear ingredient is built on the frequency-restricted estimates of [14].

Phase function and Duhamel formulation. We pass to the interaction representation by setting $v(t) := \mathcal{F}_z\big(U(-t)u(t)\big)$, where $U(t)$ is the linear propagator associated with (NV). The quadratic part of the dispersion is then encoded in

$$\Phi(\xi, \xi_1, \xi_2) = \Phi_0(\xi, \xi_1, \xi_2) + E\,\Phi_E(\xi, \xi_1, \xi_2), \qquad \xi, \xi_1, \xi_2 \in \mathbb{R}^2 \simeq \mathbb{C},$$

with

$$\Phi_0(\xi, \xi_1, \xi_2) = \mathrm{Re}(-\xi^3 + \xi_1^3 + \xi_2^3) = -3\,\mathrm{Re}(\xi\xi_1\xi_2),$$
$$\Phi_E(\xi, \xi_1, \xi_2) = -\,\mathrm{Re}\Big(-\tfrac{\xi^3}{|\xi|^2} + \tfrac{\xi_1^3}{|\xi_1|^2} + \tfrac{\xi_2^3}{|\xi_2|^2}\Big),$$

together with the bilinear multiplier $K(\xi_1, \xi_2) = \dfrac{\xi_1 \cdot \xi_2}{|\xi_1|^2|\xi_2|^2}$. Equation (NV) then reads, schematically,

$$v(t, \xi) = v(0, \xi) + \int_0^t\!\!\int_{\xi=\xi_1+\xi_2} e^{it'\Phi} K\Phi_0\, v(t', \xi_1)v(t', \xi_2)\, d\xi_1\, dt'. \tag{1.2}$$

Splitting the multiplier and time-differentiation. A short calculation gives $K\Phi_0 = K\Phi - E\,K\Phi_E$. The piece carrying $K\Phi_E$ is perturbative since Φ_E is of lower order than Φ_0. The harder piece is the one carrying $K\Phi$: it sits at the heart of the obstructions in [23, 24, 3, 1], and it is precisely the piece on which the identity $e^{it\Phi} = \partial_t e^{it\Phi}/(i\Phi)$ can be used to trade an oscillation for a power of Φ^{-1}. If applied globally this manoeuvre would create spurious singularities on the resonance set $\{\Phi = 0\}$, so we partition the convolution region into a *good* part, where the resulting symbol is benign and can be estimated directly, and a *bad* part, complementary to a neighbourhood of the resonant locus, on which division by Φ poses no problem.

Recursive expansion and gauge cancellation. The construction is then recursive: applying the same trade on the bad part produces, at each step, a residual term plus a *boundary* contribution, and feeding $\partial_t v$ back from the equation generates a new collection of terms indexed by binary trees. The boundary contributions, taken together, are the actual obstruction to closing a contraction at $s > -1$. The key observation in this paper is that the entire *series* of boundary terms can be eliminated by a single nonlinear change of variables – the gauge transform $\mathcal{G}_\delta$, defined by summing up the normal-form expansion. For small smooth solutions the gauged variable is bi-Lipschitz equivalent to the original one; for the equation it satisfies, however, the difficult quadratic interactions have been removed at the price of a hierarchy of multilinear terms whose multipliers are tractable in the frequency-restricted framework. What emerges is the gauged Novikov–Veselov equation ($\mathcal{G}_\delta$-NV),

$$\partial_t z + (\partial^3 + \overline{\partial}^3)z + E(\overline{\partial}^{-1}\partial^2 + \partial^{-1}\overline{\partial}^2)z = \sum_{j=1}^{\infty} \sum_{\mathcal{T}\in \mathrm{T}_j} \sum_{\ell=1}^{5} \mathcal{N}_\ell(\mathcal{T}; z),$$

where the sums run over binary trees of every order $j \geq 1$ and over five algebraic types of contributions $\mathcal{N}_\ell$.

Contraction for the gauged equation and the role of $s_c = -1$. For the gauged equation we run a fixed-point argument in short-time $X^{s,b}$ spaces (Proposition 4.1). The contraction closes thanks to the master multilinear estimate

$$\Big\| \int_0^t U(t-t')\mathcal{N}_\ell(\mathcal{T}; z_1, \ldots, z_{j+1})(t')\, dt' \Big\|_{X_T^{s,b}} \leq T^{0+} C^j \delta^{(2s+2)(j-3)-1} \prod_{k=1}^{j+1} \|z_k\|_{X_T^{s,b}},$$

which holds for every $j \geq 1$, every $\mathcal{T} \in \mathrm{T}_j$, and every $\ell \in \{1, \ldots, 5\}$ (Proposition 3.1). The decisive feature of this bound is the exponent $(2s + 2)(j - 3) - 1$: it is summable over j for any $s > -1$, provided the gauge parameter δ is taken small in terms of the size of the data. This is the precise point at which the scaling exponent $s_c = -1$ becomes operational in the analysis. Proposition 3.1 is in turn reduced, through the standard frequency-restricted-estimate machinery, to bounds on integrals over level sets $\{\Phi - \alpha = 0\}$, uniformly in α. Two regimes appear in the resonance geometry of Φ: a generic non-degenerate stratum, and a stationary locus on which two input frequencies are nearly aligned. Lemma 3.6 treats both, by a direct change of variables on the first and a parameter-dependent application of Morse's lemma on the second. The argument then unfolds across the five algebraic tree types and seven case-by-case configurations.

Returning to (NV): inversion of the gauge and density. To carry the well-posedness statement back to (NV) we rely on two ingredients. Lemma 4.3 establishes that $\mathcal{G}_\delta$ and its inverse are bi-Lipschitz, in fact analytic, on suitable balls of H^s. Proposition 4.5 then shows that any smooth solution of (NV) is mapped by $\mathcal{G}_\delta$ to the unique solution of ($\mathcal{G}_\delta$-NV) with matching gauged data. Smoothing the initial datum and combining the high-regularity well-posedness theory of [3, 23, 24] with uniform-in-n control on the gauged flow upgrades, by density, the smooth-level equivalence to a continuous-in-H^s statement, which is exactly Theorem 1.1.

Appendix: analyticity in the range $s > -\frac{3}{4}$. A short appendix is devoted to the (already-known) range $s > -\frac{3}{4}$. There the gauge is not needed: the same frequency-restricted estimates already yield a contraction in $X^{s,b}$ directly for (NV) when $E \neq 0$. This both recovers the result of Adams and Grünrock [1] (originally stated for $E = 0$) and promotes the data-to-solution map of Theorem 1.1 from continuous to analytic in that range.

1.3. **Organization and notation.** The remainder of the paper is structured as follows. Section 2 sets up the tree formalism underlying the infinite normal-form expansion, derives the normal-form equation for (NV), and constructs the gauge transform $\mathcal{G}_\delta$, culminating in the gauged equation ($\mathcal{G}_\delta$-NV). Section 3 introduces the Fourier restriction space $X^{s,b}$, presents the frequency-restricted estimate lemmas, and proves the master multilinear bound Proposition 3.1; the necessary input on the resonance geometry is gathered in Lemma 3.6. Section 4 develops the local theory for the gauged equation (Proposition 4.1), records the analyticity of the gauge map (Lemma 4.3), proves the equivalence between (NV) and ($\mathcal{G}_\delta$-NV) for smooth solutions (Proposition 4.5), and closes the density argument that yields Theorem 1.1. Appendix A handles the auxiliary regime $s > -\frac{3}{4}$ and refines the regularity of the data-to-solution map there.

We conclude with the notation used throughout the paper. For $A, B \in \mathbb{R}$, the symbol $A \lesssim B$ means $A \leq CB$ for some $C > 0$, while $A \sim B$ stands for the two-sided bound $A \lesssim B \lesssim A$; the slightly stronger comparability $\max(|A|, |B|) \gg \big||A| - |B|\big|$ is denoted $A \simeq B$. We write $\alpha+$ and $\alpha-$ for $\alpha + \varepsilon$ and $\alpha - \varepsilon$, respectively, with $\varepsilon > 0$ chosen sufficiently small. The plane $\mathbb{R}^2$ is identified with $\mathcal{C}$ via $\xi = (\xi_1, \xi_2) \simeq \xi_1 + i\xi_2$, and $\partial = \frac{1}{2}(\partial_x + i\partial_y)$, $\overline{\partial} = \frac{1}{2}(\partial_x - i\partial_y)$ are the corresponding Wirtinger derivatives. We use the Japanese bracket $\langle \xi \rangle := (1+|\xi|^2)^{1/2}$, and we denote by $\mathcal{F}$ and $\mathcal{F}^{-1}$ (or $\widehat{\cdot}$) the spatial Fourier transform and its inverse on $\mathbb{R}^2$. The phase symbol attached to the quadratic part of (NV) is $\Phi = \Phi_0 + E\,\Phi_E$, with Φ_0 and Φ_E as in Section 1.2, and $K(\xi_1, \xi_2) = \xi_1 \cdot \xi_2 / (|\xi_1|^2 |\xi_2|^2)$ is the bilinear multiplier appearing in (1.2). For $j \geq 1$, we let T_j stand for the set of decorated binary trees of order j defined in Section 2, and we write $\mathcal{N}_\ell(\mathcal{T}; \cdot)$ for the multilinear operator associated with a tree $\mathcal{T}$ and a type label $\ell \in \{1, \dots, 5\}$. The usual Sobolev and Schwartz spaces are $H^s(\mathbb{R}^2)$, $\mathcal{S}(\mathbb{R}^2)$, $\mathcal{S}'(\mathbb{R}^2)$; finally, $X^{s,b} = X^{s,b}(\mathbb{R} \times \mathbb{R}^2)$ denotes the Bourgain-type Fourier restriction space adapted to the dispersion symbol of (NV), introduced in Section 3.

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2. Preliminaries

2.1. **Tree formalism.**

Definition 2.1. A *binary tree* $\mathcal{T}$ is a partially ordered set (with partial order $\leq$) satisfying the following properties:

(i) $\mathcal{T}$ has a unique minimal element called the *root node* labelled by $\emptyset$ (i.e., $\emptyset \leq a$ for all $a \in \mathcal{T}$).

(ii) Given $a \in \mathcal{T} \setminus \{\emptyset\}$, there exists a unique $b \in \mathcal{T}$ such that: $b \neq a$, $b \leq a$, and $b \leq c \leq a$ implies $c \in \{a, b\}$. In that case, we say that b is the *parent* of a and a is a *child* of b.

(iii) A node $a \in \mathcal{T}$ is called *terminal* or a *leaf* if it has no children. A *non-terminal node* or a *parent* $a \in \mathcal{T}$ is a node with exactly two children, one called *left child* and the other *right child*.[1] We write $\mathcal{T}^0, \mathcal{T}^\infty$ for the sets of parent and terminal nodes of $\mathcal{T}$, respectively. Note that $\mathcal{T}$ is the disjoint union of $\mathcal{T}^0$ and $\mathcal{T}^\infty$.

[1] Introducing the notation of left and right children of a node reflects the lack of ambiguity in the usual planar representation of binary trees.

(iv) Each node in $\mathcal{T}$ is labelled with a word in the following way: the root node has word $\emptyset$; and for a given parent node $a \in \mathcal{T}^0$, its left and right children have labels $a1$ and $a2$, obtained by concatenating 1 or 2 to the right of a, respectively. Note that children of the root node have labels 1 and 2.

We interchangeably use $\mathcal{T}$ to denote the representation of the tree as a finite connected graph and to denote the set of words used to label its nodes. Note that the latter is also a partially ordered set, with words a, b satisfying $a \leq b$ if there exists a word c such that $b = ac$.

Given $j \in \mathbb{N}$, we use the notation T_j to denote the family of all binary trees with j non-terminal nodes, which we call the j-th generation of trees. From Definition 2.1, it follows that

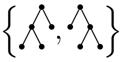
$$|\mathcal{T}^0| = j \quad \text{and} \quad |\mathcal{T}^\infty| = j + 1.$$

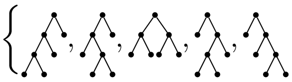
It is not difficult to see that the number of distinct trees in T_j is given by the j-th Catalan number

$$|\mathrm{T}_j| = \frac{1}{j+1}\binom{2j}{j} = \frac{(2j)!}{(j+1)!j!}. \tag{2.1}$$

In particular, the first 3 generations of trees are succintly written below, identified by their graphs:

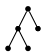
$\mathrm{T}_1 = \{$  $\}$,

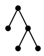
$\mathrm{T}_2 = \{$  $\}$,

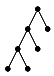
$\mathrm{T}_3 = \{$  $\}$.

Similarly, we can represent the trees above by the following partially ordered sets of words used to label their nodes (as per Definition 2.1(iv)):

 $= \{\emptyset\}$,

 $= \{\emptyset, 1, 2, 11, 12\}$,

 $= \{\emptyset, 1, 2, 21, 22\}$,

 $= \{\emptyset, 1, 2, 11, 12, 111, 112\}, \ldots$

For readibility, we will reserve the symbols $\bullet$, $\star$, $\circ$, and $*$ to denote nodes in a tree and their respective associated word. Next, we introduce some additional notation to identify different nodes in a tree.

Definition 2.2. Let $\mathcal{T}$ be a binary tree in the sense of Definition 2.1 and $\bullet, \star$ be two nodes in $\mathcal{T}$.

(i) If $\bullet \in \mathcal{T} \setminus \{\emptyset\}$, we write $\mathbf{p}(\bullet)$ to denote the *parent node* of $\bullet$. If $\mathbf{p}(\bullet) \neq \emptyset$, we write $\mathbf{g}(\bullet) = \mathbf{p}(\mathbf{p}(\bullet))$.

(ii) If $\bullet, \star \in \mathcal{T} \setminus \{\emptyset\}$ are distinct and share the same parent, they are called *siblings*. We write $\mathbf{s}(\bullet)$ for the sibling of $\bullet$.

(iii) Given $\bullet \in \mathcal{T}^0$, we say $\bullet$ is a *final parent* if its children $\bullet 1, \bullet 2 \in \mathcal{T}^\infty$. The set of final parents is denoted by $\mathcal{T}^{0,\mathfrak{f}}$.

Remark 2.3. Note that from Definition 2.1, the maps $\mathbf{p}, \mathbf{s}$, for the parent and sibling of a given node, are well-defined in their respective domains, since if either the parent or sibling node exists, it is unique.

We next introduce concatenation and subtraction of trees.

Definition 2.4. Let $\mathcal{T}$ be a binary tree in the sense of Definition 2.1.

(i) (Concatenation) Given $\star \in \mathcal{T}^\infty$, we define

$$\mathcal{T} \overset{\star}{\oplus} \bigwedge := \mathcal{T} \cup \{\star 1, \star 2\}$$

as the result of adding the children $\star 1, \star 2$ of $\star$ to the tree $\mathcal{T}$. Note that this operation now makes $\star$ a final parent in the tree $\mathcal{T} \overset{\star}{\oplus} \bigwedge$.

(ii) (Subtraction) Given a final parent $\star \in \mathcal{T}^{0,\mathfrak{f}}$, we define the tree

$$\mathcal{T} \overset{\star}{\ominus} \bigwedge := \mathcal{T} \setminus \{\star 1, \star 2\}$$

as the result of eliminating the children $\star 1, \star 2$ of $\star$ from $\mathcal{T}$. Note that $\star$ becomes a terminal node in the new tree $\mathcal{T} \overset{\star}{\ominus} \bigwedge$. More generally, given $C \subseteq \mathcal{T}^{0,f}$, we define

$$\mathcal{T} \overset{C}{\ominus} \bigwedge := \bigcap_{\star \in C} (\mathcal{T} \overset{\star}{\ominus} \bigwedge) = \bigcap_{\star \in C} (\mathcal{T} \setminus \{\star 1, \star 2\}) = \mathcal{T} \setminus \Big(\bigcup_{\star \in C} \{\star 1, \star 2\} \Big),$$

which corresponds to the set where all the children of the final parents $\star \in C$ have been removed from the tree.

The tree structure defined above and its word-labelling will help in keeping track of the nonlinear terms appearing in the normal form reduction performed in Subsection 2.2. There, after a differentiation-by-parts step in time, a time derivative falls on a v-factor and is replaced, via the equation for $\partial_t \widehat{v}$, by a quadratic term, generating a higher-degree nonlinear contribution. Binary trees provide a natural bookkeeping for the frequencies of the corresponding factors, in the following way.

Definition 2.5. Let $j \in \mathbb{N}$ and $\mathcal{T} \in \mathrm{T}_j$.

(i) (Index function) We define the index function $\vec{\xi} : \mathcal{T} \to \mathbb{C}^{\mathcal{T}}$ such that $\vec{\xi} = (\xi_\bullet)_{\bullet \in \mathcal{T}}$, where

$$\xi_\bullet = \xi_{\bullet 1} + \xi_{\bullet 2}, \quad \bullet \in \mathcal{T}^0,$$

and $\bullet 1$ and $\bullet 2$ denote the children of $\bullet$. Moreover, we define the set $\Gamma(\mathcal{T})$ as the set of convolution relations on the tree $\mathcal{T}$:

$$\Gamma(\mathcal{T}) := \big\{ \vec{\xi} \in \mathbb{C}^{\mathcal{T}} : \ \xi_\bullet = \xi_{\bullet 1} + \xi_{\bullet 2}, \ \bullet \in \mathcal{T}^0 \big\}.$$

(ii) (Zero-energy phase function) We use $\Psi_0(\mathcal{T})$ to denote the *zero-energy phase function* associated with the tree $\mathcal{T}$, given by

$$\Psi_0(\mathcal{T}) := \mathrm{Re} \left(-\xi^3 + \sum_{\bullet \in \mathcal{T}^\infty} \xi_\bullet^3 \right) \tag{2.2}$$

where $\mathcal{T}^\infty$ denotes the set of leaves of $\mathcal{T}$.

Given a parent node $\star \in \mathcal{T}^0$, we also introduce the partial zero-energy phase function for the tree $\overset{\star}{\bigwedge} = \{\star, \star 1, \star 2\}$:

$$\Psi_0(\star) = \Psi_0(\overset{\star}{\bigwedge}) := -\xi_\star^3 + \xi_{\star 1}^3 + \xi_{\star 2}^3. \tag{2.3}$$

Note that the following additivity property holds for Ψ:

$$\Psi_0(\mathcal{T}) = \sum_{\star \in \mathcal{T}^0} \Psi_0(\star). \tag{2.4}$$

(iii) (Energy phase function) Given a binary tree $\mathcal{T}$, we also define the *energy phase function* as

$$\Psi_E(\mathcal{T}) := -\operatorname{Re}\left(-\frac{\xi^3}{|\xi|^2} + \sum_{\bullet\in\mathcal{T}^\infty} \frac{\xi_\bullet^3}{|\xi_\bullet|^2}\right).$$

The partial energy phase function is given as

$$\Psi_E(\star) = \Psi_E(\overset{\star}{\bigwedge}).$$

(iv) (Phase function) For a fixed binary tree $\mathcal{T}$, the (total) phase function is then defined as

$$\Psi(\mathcal{T}) = \Psi_0(\mathcal{T}) + E\Psi_E(\mathcal{T}).$$

Given $\star \in \mathcal{T}^0$, we write

$$\Psi(\star) = \Psi_0(\star) + E\Psi_E(\star)$$

(v) Lastly, for $\star \in \mathcal{T}^0$, we define the multiplier $K_\star$ given by

$$K_\star := \frac{\xi_{\star 1}\cdot\xi_{\star 2}}{|\xi_{\star 1}|^2|\xi_{\star 2}|^2}, \tag{2.5}$$

where $\xi_{\star 1}\cdot\xi_{\star 2}$ is the $\mathbb{R}^2$ inner product.

To describe the infinite normal form reduction, we introduce a correspondence between binary trees and oscillatory integrals. Given $j \in \mathbb{N}$, a binary tree $\mathcal{T} \in \mathrm{T}_j$, a multiplier $m : \mathbb{C}^{\mathcal{T}} \to \mathbb{C}$ on $\mathcal{T}$, and a function $v : \mathbb{R}\times\mathbb{C} \to \mathbb{C}$, we define the associated integral term

$$\mathcal{I}(\mathcal{T}, m; v)(t) := i\int_{\Gamma(\mathcal{T})} e^{it\Psi(\mathcal{T})} m(\vec{\xi}) \prod_{\bullet\in\mathcal{T}^\infty} \widehat{v}_\bullet(t)d\vec{\xi}, \tag{2.6}$$

where we use the short-hand notation $\widehat{v}_\bullet(t) := \widehat{v}(t, \xi_\bullet)$ for a label $\bullet$.

2.2. **Deriving the normal form equation.** Using the tree formalism in Subsection 2.1, we now derive the normal form equation which we shall use. We apply a normal form reduction in the form of "differentiation-by-parts" as in [4], only in a problematic region where we cannot show the nonlinear estimate at low regularity. In particular, we consider the following sets A and $A^c = \Gamma(\bigwedge) \setminus A$:

$$\begin{aligned} A &:= \big\{(\xi_1, \xi_2) \in \mathbb{R}^2 :\ \xi_1 + \xi_2 = \xi,\ |\xi_1| \vee |\xi_2| \le 1/\delta\big\}, \\ A^c &:= \big\{(\xi_1, \xi_2) \in \mathbb{R}^2 :\ \xi_1 + \xi_2 = \xi,\ |\xi_1| \vee |\xi_2| \ge 1/\delta\big\}, \end{aligned} \tag{2.7}$$

where A^c denotes the bad region where we perform a reduction.

Using the tree-indexing in Subsection 2.1 and the notations in (2.2), (2.3), and (2.5), the equation satisfied by $v = \mathcal{F}_z(U(-t)u)$, in the interaction representation, can be written as follows:

$$\partial_t\widehat{v}(t) = i\int_{\xi=\xi_1+\xi_2} e^{it\Psi(\bigwedge)} K\Psi_0(\bigwedge)(\widehat{v}_1\widehat{v}_2)(t)d\xi_1. \tag{2.8}$$

(Absolute multiplicative constants in front of the integral have been absorbed into the multiplier; see the discussion preceding (1.2).) Then, splitting the region of integration into $A \cup A^c$ as in (2.7), we may write (2.8) as

$$\partial_t\widehat{v} = \mathcal{I}\big(\bigwedge, \mathbb{1}_A K\Psi_0(\bigwedge); v\big) + \mathcal{I}\big(\bigwedge, \mathbb{1}_{A^c} K\Psi_0(\bigwedge); v\big)$$

where $\mathcal{I}$ is as in (2.6). In order to perform a differentiation-by-parts in time on the M-term restricted to the bad region A^c, we need to make the total phase $\Psi(\bigwedge)$ appear. Hence, we simply recall that

$$\Psi(\bigwedge) = \Psi_0(\bigwedge) + E\Psi_E(\bigwedge).$$

This allows us to write

$$\begin{aligned}\partial_t\widehat{v} &= \mathcal{I}\big(\bigwedge, \mathbb{1}_A K\Psi_0(\bigwedge); v\big) + \mathcal{I}\big(\bigwedge, \mathbb{1}_{A^c} K\Psi(\bigwedge); v\big)\\ &\quad - \mathcal{I}\big(\bigwedge, E\cdot K\cdot \Psi_E(\bigwedge); v\big) + \mathcal{I}\big(\bigwedge, \mathbb{1}_A K\cdot E\cdot \Psi_E(\bigwedge); v\big)\\ &= \mathcal{I}\big(\bigwedge, \mathbb{1}_A K\Psi_0(\bigwedge) - \mathbb{1}_{A^c} K\cdot E\cdot\Psi_E(\bigwedge); v\big) + \mathcal{I}(\bigwedge, \mathbb{1}_{A^c} K\Psi(\bigwedge); v)\\ &=: \mathcal{F}_x[\mathrm{N}(\bigwedge; v)] + \mathcal{F}_x[\mathrm{M}(\bigwedge; v)]. \end{aligned} \tag{2.9}$$

In order to move on with our development, we start by using the relation

$$\partial_t \frac{e^{it\Psi(\bigwedge)}}{i\Psi(\bigwedge)} = e^{it\Psi(\bigwedge)},$$

which allows us to perform the first differentiation-by-parts step in our chain. We obtain

$$\begin{aligned}\partial_t\widehat{v}(t) &= \mathcal{F}_x[\mathrm{N}(\bigwedge; v)](t) + \partial_t\Big[\int_{\xi=\xi_1+\xi_2} e^{it\Psi(\bigwedge)}[\mathbb{1}_{A^c}K](\widehat{v}_1\widehat{v}_2)(t)d\xi_1\Big]\\ &\quad - \int_{\xi=\xi_1+\xi_2} e^{it\Psi(\bigwedge)}[\mathbb{1}_{A^c}K]\partial_t(\widehat{v}_1\widehat{v}_2)(t)d\xi_1\\ &= \mathcal{F}_x[\mathrm{N}(\bigwedge; v)](t) + \partial_t\Big[\int_{\xi=\xi_1+\xi_2} e^{it\Psi(\bigwedge)}[\mathbb{1}_{A^c}K](\widehat{v}_1\widehat{v}_2)(t)d\xi_1\Big]\\ &\quad - i\int_{\substack{\xi=\xi_1+\xi_2\\ \xi_1=\xi_{11}+\xi_{12}}} \exp\Big(it\Psi({\scriptstyle\bigwedge}\!\bigwedge)\Big)[\mathbb{1}_{A^c}KK_1\Psi_0(1)](\widehat{v}_{11}\widehat{v}_{12}\widehat{v}_2)(t)d\xi_1 d\xi_{11}\\ &\quad - i\int_{\substack{\xi=\xi_1+\xi_2\\ \xi_2=\xi_{21}+\xi_{22}}} \exp\Big(it\Psi(\bigwedge\!{\scriptstyle\bigwedge})\Big)[\mathbb{1}_{A^c}KK_2\Psi_0(2)](\widehat{v}_1\widehat{v}_{21}\widehat{v}_{22})(t)d\xi_1 d\xi_{21}\end{aligned} \tag{2.10}$$

where we replaced $\partial_t\widehat{v}_j$ by (2.8) to obtain the last equality. This can be rewritten as

$$\begin{aligned}\partial_t\widehat{v} &= \mathcal{F}_x[\mathrm{N}(\bigwedge; v)] + \partial_t\mathcal{I}\big(\bigwedge, -i\mathbb{1}_{A^c}K; v\big)\\ &\quad + \mathcal{I}\Big({\scriptstyle\bigwedge}\!\bigwedge, -\mathbb{1}_{A^c}KK_1\Psi_0(1); v\Big) + \mathcal{I}\Big(\bigwedge\!{\scriptstyle\bigwedge}, -\mathbb{1}_{A^c}KK_2\Psi_0(2); v\Big).\end{aligned} \tag{2.11}$$

In order to iterate the procedure taking (2.8) into (2.11), we use the following result. It provides a systematic way to perform a normal form reduction on integral terms of the form $\mathcal{I}$ in (2.6), which translates to an operation on trees (based on tree concatenation as defined in Definition 2.4 (ii)) and suitable modifications of the multipliers. We refer the reader to the companion KdV paper [11] for the proof, which works in the exact same fashion in our setting.

Lemma 2.6 (Differentiation-by-parts on trees)**.** *Let $j \in \mathbb{N}$, $\mathcal{T} \in \mathrm{T}_j$, $m : \mathbb{R}^{\mathcal{T}} \to \mathbb{C}$ a multiplier on $\mathcal{T}$, and v be a smooth function such that $\widehat{v}$ satisfies* (2.8)*. Then, the following identity holds*

$$\mathcal{I}(\mathcal{T}, m; v) = \partial_t\mathcal{I}\Big(\mathcal{T}, \frac{m}{i\Psi(\mathcal{T})}; v\Big) + \sum_{\star\in\mathcal{T}^\infty}\mathcal{I}\Big(\mathcal{T}\overset{\star}{\oplus}\bigwedge, -K_\star\Psi_0(\star)\frac{m}{\Psi(\mathcal{T})}; v\Big), \tag{2.12}$$

where $\mathcal{I}$ is the integral functional in (2.6)*, $\mathcal{T}\overset{\star}{\oplus}\bigwedge$ denotes tree concatenation at the final node $\star\in\mathcal{T}^\infty$ as in Definition 2.4 (ii), and $\Psi(\mathcal{T})$, $\Psi(\star)$, $K_\star$ are as in* (2.2)*,* (2.3)*, and* (2.5)*.*

As in (2.10), we split each region of frequency integration between a good and a bad set, and only perform a differentiation-by-parts when restricted to the latter. Generalizing the definition of A, A^c in (2.7), given a tree $\mathcal{T}$ and a parent node $\bullet \in \mathcal{T}^0$, we define the analogous sets $A_\bullet$ and $A_\bullet^c$ as

$$\begin{aligned} A_\bullet &:= \big\{(\xi_{\bullet 1}, \xi_{\bullet 2}) \in \mathbb{R}^2 : \ \xi_\bullet = \xi_{\bullet 1} + \xi_{\bullet 2}, \ |\xi_{\bullet 1}| \vee |\xi_{\bullet 2}| \le 1/\delta\big\}, \\ A_\bullet^c &:= \big\{(\xi_{\bullet 1}, \xi_{\bullet 2}) \in \mathbb{R}^2 : \ \xi_\bullet = \xi_{\bullet 1} + \xi_{\bullet 2}, \ |\xi_{\bullet 1}| \vee |\xi_{\bullet 2}| \ge 1/\delta\big\}. \end{aligned}$$

Note that, when $\bullet = \emptyset$ is the root node, $A_\bullet = A$ as in (2.7). We will often omit the dependence on the children frequencies $\xi_{\bullet 1}, \xi_{\bullet 2}$ of $\xi_\bullet$, and use the short-hand notation $\mathbb{1}_{A_\bullet} = \mathbb{1}_{A_\bullet}(\xi_{\bullet 1}, \xi_{\bullet 2})$.

Returning to (2.11), we continue to derive the second iteration of the normal form equation. Unlike the first normal form reduction from equation (2.8) to (2.11), where we split the integral term into A and A^c (see (2.7)) and differentiate-by-parts in the latter region, we now have an additional step, which we describe in the following. Considering the two last terms in (2.11), we first split their regions of integration $(\xi_{\bullet 1}, \xi_{\bullet 2}) \in A_\bullet \cup A_\bullet^c$, where $\bullet = 1, 2$, respectively:

$$\begin{aligned} \partial_t \widehat{v} &= \mathcal{F}_x[\mathrm{N}(\bigwedge; v)] + \partial_t \mathcal{I}\big(\bigwedge, -i\mathbb{1}_{A^c} K; v\big) \\ &\quad + \mathcal{I}\Big({}^{\wedge}\!\!\bigwedge, -\mathbb{1}_{A^c \cap A_1} K K_1 \Psi_0(1); v\Big) + \mathcal{I}\Big({}^{\wedge}\!\!\bigwedge, -\mathbb{1}_{A^c \cap A_1^c} K K_1 \Psi_0(1); v\Big) \\ &\quad + \mathcal{I}\Big(\bigwedge\!\!{}^{\wedge}, -\mathbb{1}_{A^c \cap A_2} K K_2 \Psi_0(2); v\Big) + \mathcal{I}\Big(\bigwedge\!\!{}^{\wedge}, -\mathbb{1}_{A^c \cap A_2^c} K K_2 \Psi_0(2); v\Big). \end{aligned}$$

As a first preparatory step, we add and subtract the 'energy' terms, in order to complete the phases in the last couple of lines above. Indeed, we first write

$$\Psi(\star) = \big(\Psi_0(\star) + E \cdot \Psi_E(\star)\big) - E \cdot \Psi_E(\star),$$

which allows us to write

$$\begin{aligned} \partial_t \widehat{v} &= \mathcal{F}_x[\mathrm{N}(\bigwedge; v)] + \partial_t \mathcal{I}\big(\bigwedge, -i\mathbb{1}_{A^c} K; v\big) \\ &\quad + \mathcal{I}\Big({}^{\wedge}\!\!\bigwedge, K K_1 \mathbb{1}_{A^c} \cdot \big(E \mathbb{1}_{A_1^c} \Psi_E(1) - \mathbb{1}_{A_1} \Psi_0(1)\big)\,; v\Big) + \mathcal{I}\Big({}^{\wedge}\!\!\bigwedge, -\mathbb{1}_{A^c \cap A_1^c} K K_1 \Psi(1); v\Big) \\ &\quad + \mathcal{I}\Big(\bigwedge\!\!{}^{\wedge}, K K_2 \mathbb{1}_{A^c} \big(E \mathbb{1}_{A_2^c} \Psi_E(2) - \mathbb{1}_{A_2} \Psi_0(2)\big)\,; v\Big) + \mathcal{I}\Big(\bigwedge\!\!{}^{\wedge}, -\mathbb{1}_{A^c \cap A_2^c} K K_2 \Psi(2); v\Big). \end{aligned}$$

As before, we keep the integral contributions which either come from a 'lower-order' or energy term, or which are localized to a "good set" $A_\bullet$. Regarding the contributions restricted to $A_\bullet^c$ for all $\bullet \in \mathcal{T}^0$, despite the restriction to the "bad regions", we will keep a suitable part of these terms before performing another differentiation-by-parts. In particular, by using the additivity of the phase Ψ in (2.4), we write

$$\Psi(\star) = \big[\Psi(\star) + \Psi(\bigwedge)\big] - \Psi(\bigwedge) = \Psi(\mathcal{T}) - \Psi(\bigwedge), \quad (\star, \mathcal{T}) \in \Big\{(1, {}^{\wedge}\!\!\bigwedge), (2, \bigwedge\!\!{}^{\wedge})\Big\},$$

and keep the contributions with partial phase $-\Psi(\bigwedge)$, only differentiating-by-parts the terms with total phase $\Psi(\mathcal{T})$. Then, from (2.9), we obtain the second iteration of the normal form equation

$$
\begin{aligned}
\partial_t \widehat{v} &= \mathcal{F}_x[\mathrm{N}(\bigwedge; v)] + \partial_t \mathcal{I}\big(\bigwedge, -i\mathbb{1}_{A^c}K; v\big) \\
&\quad + \mathcal{I}\Big(\bigwedge^{\!\bullet}_{\;\bullet\bullet}, KK_1\mathbb{1}_{A^c}\cdot\big(\mathbb{1}_{A_1^c}\Psi(\bigwedge) + E\mathbb{1}_{A_1^c}\Psi_E(1) - \mathbb{1}_{A_1}\Psi_0(1)\big)\,; v\Big) \\
&\quad + \mathcal{I}\Big(\bigwedge^{\!\bullet}_{\;\bullet\bullet}, -\mathbb{1}_{A^c\cap A_1^c}KK_1\Psi(\bigwedge^{\!\bullet}_{\;\bullet\bullet}); v\Big) \\
&\quad + \mathcal{I}\Big({}^{\bullet}\!\bigwedge_{\bullet\bullet}, KK_2\mathbb{1}_{A^c}\,\big(\mathbb{1}_{A_2^c}\Psi(\bigwedge) + E\mathbb{1}_{A_2^c}\Psi_E(2) - \mathbb{1}_{A_2}\Psi_0(2)\big)\,; v\Big) \\
&\quad + \mathcal{I}\Big({}^{\bullet}\!\bigwedge_{\bullet\bullet}, -\mathbb{1}_{A^c\cap A_2^c}KK_2\Psi({}^{\bullet}\!\bigwedge_{\bullet\bullet}); v\Big) \\
&=: \mathcal{F}_x[\mathrm{N}(\bigwedge; v)] + \mathcal{F}_x[\mathrm{B}(\bigwedge; v)] \\
&\qquad + \mathcal{F}_x[\mathrm{N}(\bigwedge^{\!\bullet}_{\;\bullet\bullet}; v)] + \mathcal{F}_x[\mathrm{M}(\bigwedge^{\!\bullet}_{\;\bullet\bullet}; v)] + \mathcal{F}_x[\mathrm{N}({}^{\bullet}\!\bigwedge_{\bullet\bullet}; v)] + \mathcal{F}_x[\mathrm{M}({}^{\bullet}\!\bigwedge_{\bullet\bullet}; v)].
\end{aligned}
\tag{2.13}
$$

In the Duhamel expansion of v, based on (2.13), N corresponds to time-integral terms, B to boundary terms, and M are remainders, to which we apply (2.12) at the next step. Thus, in the following, we refer to them as integral terms, boundary terms, and remainders, respectively.

In the following iterations of the normal form equation, as in (2.11) and (2.13), for each tree $\mathcal{T}$, we want to have a unique integral term $\mathrm{N}(\mathcal{T})$ and boundary term $\mathrm{B}(\mathcal{T})$, where this must encode all the possible ways of arriving at such a tree structure. From Lemma 2.6, we note that given a tree $\mathcal{T} \in \mathrm{T}_j$ for $j \geq 2$, after differentiation-by-parts, the resulting contributions associated with the tree $\mathcal{T}$ can come from any term of the form $\mathcal{I}(\mathcal{T} \overset{\star}{\ominus} \bigwedge)$ where $\star \in \mathcal{T}^{0,\mathfrak{f}}$, i.e., that the children of $\star$ are leaves in $\mathcal{T}$. Consequently, in the following, we need to perform differentiation-by-parts on the remainder terms $\mathrm{M}(\mathcal{T})$ simultaneously and combine any contributions associated with the same tree before preparing the integral terms to be kept and those which will be differentiated in the following step.

The following result provides a systematic way of repeating the process described above, with an explicit description of the contributions and multipliers involved at each finite normal form equation. We note that we see one boundary term and an integral term for each tree of a given generation T_j, with the complexity of the contributions being encoded in the corresponding multipliers.

Proposition 2.7. *Let $J \in \mathbb{N}$, u be a smooth solution to* (NV)*, and $v = \mathcal{F}_z(U(-t)u)$ be the corresponding solution to* (2.8)*. Then, the J-th normal form equation for v is given by*

$$
\partial_t v = \sum_{j=1}^{J-1} \sum_{\mathcal{T} \in \mathrm{T}_j} \mathrm{B}(\mathcal{T}; v) + \sum_{j=1}^{J} \sum_{\mathcal{T} \in \mathrm{T}_j} \mathrm{N}(\mathcal{T}; v) + \sum_{\mathcal{T} \in \mathrm{T}_J} \mathrm{M}(\mathcal{T}; v),
$$

where the operators are defined via their spatial Fourier transforms

$$
\begin{aligned}
\mathcal{F}_x[\mathrm{B}(\mathcal{T}; v)] &= \partial_t \mathcal{I}(\mathcal{T}, -i\,\mathrm{mul}_{\mathrm{B}}(\mathcal{T}); v), \\
\mathcal{F}_x[\mathrm{N}(\mathcal{T}; v)] &= \mathcal{I}(\mathcal{T}, \mathrm{mul}(\mathcal{T}); v), \\
\mathcal{F}_x[\mathrm{M}(\mathcal{T}; v)] &= \mathcal{I}(\mathcal{T}, \Psi(\mathcal{T})\,\mathrm{mul}_{\mathrm{B}}(\mathcal{T}); v),
\end{aligned}
\tag{2.14}
$$

with $\mathcal{I}(\mathcal{T}, m; v)$ *as in* (2.6). *Moreover, the multipliers* $\mathrm{mul}(\mathcal{T})$ *and* $\mathrm{mul}_{\mathrm{B}}(\mathcal{T})$ *for* $\mathcal{T} \in \mathrm{T}_j$ *are given by*

$$
\begin{aligned}
\mathrm{mul}_{\mathrm{B}}(\mathcal{T}) &= (-1)^{j+1} \prod_{\bullet \in \mathcal{T}^0} \mathbb{1}_{A^c_\bullet} K_\bullet, \\
\mathrm{mul}(\mathcal{T}) &= (-1)^{j+1} \Big(\prod_{\bullet \in \mathcal{T}^0} K_\bullet \Big) \Big\{ \sum_{\star \in \mathcal{T}^{0,f}} \big(\mathbb{1}_{A_\star} \Psi_0(\star) - \mathbb{1}_{A^c_\star} E \Psi_E(\star) \big) \Big(\prod_{\bullet \in \mathcal{T}^0 \setminus \{\star\}} \mathbb{1}_{A^c_\bullet} \Big) \\
&\qquad\qquad - \Big(\prod_{\bullet \in \mathcal{T}^0} \mathbb{1}_{A^c_\bullet} \Big) \sum_{\star \in \mathcal{T}^0 \setminus \mathcal{T}^{0,f}} \Psi(\star) \Big\},
\end{aligned}
\tag{2.15}
$$

where $K_\bullet$ *and* $\Psi(\star)$ *are as in* (2.5) *and* (2.3).

Taking $J \to \infty$ in Proposition 2.7 yields, at the formal level, the *infinite normal form equation*

$$
\partial_t v = \sum_{j=1}^{\infty} \sum_{\mathcal{T} \in \mathrm{T}_j} \left(\mathrm{N}(\mathcal{T}; v) + \mathrm{B}(\mathcal{T}; v) \right), \tag{2.16}
$$

with $\mathrm{N}(\mathcal{T}; v)$ and $\mathrm{B}(\mathcal{T}; v)$ as in (2.14).

Our route from (2.16) to the gauged equation has three stages. First, a single nonlinear substitution $v \mapsto w$ is engineered to absorb the quadratic boundary term in (2.16); the surprising fact is that the same substitution eliminates *every* boundary term via an algebraic identity (Lemma 2.8). Second, an analogous cancellation on the nonlinear side reduces the surviving multipliers to five families, which we tabulate (Lemma 2.10). Third, the interaction representation $z(t) := U(-t)w(t)$ recasts the system as the gauged Novikov–Veselov equation ($\mathcal{G}_\delta$-NV). The algebra runs in close parallel with [11], with one structural novelty: alongside the dispersive phase Ψ_0, the energy phase $E\Psi_E$ contributes its own family of multipliers (Type III), so the inclusion-exclusion bookkeeping is carried out on the pair $(\Psi_0, E\Psi_E)$ rather than on a single phase.

The multilinear functional and the change of variables. It will be convenient to evaluate $\mathcal{I}$ from (2.6) at distinct inputs. Fix $j \in \mathbb{N}$ and $\mathcal{T} \in \mathrm{T}_j$, and order the $j+1$ leaves of $\mathcal{T}$ lexicographically by their words,

$$
\bullet_1 \leq \bullet_2 \leq \cdots \leq \bullet_{j+1}, \tag{2.17}
$$

i.e. left-to-right in any planar drawing. For arbitrary smooth $v_1, \ldots, v_{j+1} : \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}$, set

$$
\mathcal{I}(\mathcal{T}, m; v_1, \ldots, v_{j+1})(t) = i \int_{\Gamma_{\mathcal{T}}} e^{it\Psi(\mathcal{T})} m(\vec{\xi}) \prod_{\ell=1}^{j+1} \widehat{v}_\ell(t, \xi_{\bullet_\ell}) \, d\vec{\xi}.
$$

Specialising $v_1 = \cdots = v_{j+1} = v$ recovers $\mathcal{I}(\mathcal{T}, m; v)$ of (2.6). The multilinear counterparts of $\mathrm{B}(\mathcal{T})$ and $\mathrm{N}(\mathcal{T})$ from (2.14) are then

$$
\begin{aligned}
\mathcal{F}_x[\mathrm{B}(\mathcal{T}; v_1, \ldots, v_{j+1})] &= \partial_t \mathcal{I}(\mathcal{T}, -i \, \mathrm{mul}_{\mathrm{B}}(\mathcal{T}); v_1, \ldots, v_{j+1}), \\
\mathcal{F}_x[\mathrm{N}(\mathcal{T}; v_1, \ldots, v_{j+1})] &= \mathcal{I}(\mathcal{T}, \mathrm{mul}(\mathcal{T}); v_1, \ldots, v_{j+1}).
\end{aligned}
$$

Define $v \mapsto w$ implicitly by

$$
v = w + \mathcal{I}(\bigwedge, -i \, \mathrm{mul}_{\mathrm{B}}(\bigwedge); w) =: w + \mathrm{D}[w, w], \tag{2.18}
$$

with $\mathcal{I}$ as in (2.6). By design, $\partial_t \mathrm{D}[w, w]$ reproduces precisely the quadratic boundary contribution $\mathrm{B}(\bigwedge; w)$, so

$$
\partial_t v = \partial_t w + \mathrm{B}(\bigwedge; w), \tag{2.19}
$$

with B as in (2.14) (cf. (2.13)). Plugging (2.19) and (2.18) into (2.16) gives the closed equation

$$
\begin{aligned}
\partial_t w &= -\mathrm{B}(\Lambda; w) + \mathrm{B}(\Lambda; v) + \mathrm{N}(\Lambda; v) + \sum_{j=2}^{\infty} \sum_{\mathcal{T}\in \mathrm{T}_j} \Big(\mathrm{N}(\mathcal{T}; v) + \mathrm{B}(\mathcal{T}; v) \Big) \\
&= \mathrm{B}\big(\Lambda; w, \mathrm{D}[w,w]\big) + \mathrm{B}\big(\Lambda; \mathrm{D}[w,w], w\big) + \mathrm{B}\big(\Lambda; \mathrm{D}[w,w], \mathrm{D}[w,w]\big) \\
&\quad + \mathrm{N}(\Lambda; w, w) + \mathrm{N}\big(\Lambda; w, \mathrm{D}[w,w]\big) + \mathrm{N}\big(\Lambda; \mathrm{D}[w,w], w\big) + \mathrm{N}\big(\Lambda; \mathrm{D}[w,w], \mathrm{D}[w,w]\big) \\
&\quad + \sum_{j=2}^{\infty} \sum_{\mathcal{T}\in \mathrm{T}_j} \Big(\mathrm{N}\big(\mathcal{T}; w + \mathrm{D}[w,w]\big) + \mathrm{B}\big(\mathcal{T}; w + \mathrm{D}[w,w]\big) \Big).
\end{aligned} \tag{2.20}
$$

Because $\mathrm{D}[w,w]$ is itself quadratic in w, expanding (2.20) produces, for each underlying tree, several terms of the same homogeneity but supported on different colourings. Bundling such terms tree-by-tree shifts the combinatorial overhead into the multipliers and reveals a striking algebraic phenomenon: *every* boundary term is annihilated.

The boundary cancellation.

Lemma 2.8. *Let $j \in \mathbb{N}$, $\mathcal{T} \in \mathrm{T}_j$, $\star \in \mathcal{T}^{0,\mathfrak{f}}$, and let $w_1, \dots, w_{j+1} : \mathbb{R}\times\mathbb{R}^2 \to \mathbb{R}$ be smooth. Order the leaves of $\mathcal{T}$ as in* (2.17)*, and let $1 \le k \le j$ be such that $(\bullet_k, \bullet_{k+1})$ are the children of $\star$ in $\mathcal{T}$. Then*

$$
\mathrm{B}\big(\mathcal{T}; w_1, \dots, w_{j+1}\big) + \mathrm{B}\big(\mathcal{T} \overset{\star}{\ominus} \Lambda; w_1, \dots, w_{k-1}, \mathrm{D}[w_k, w_{k+1}], w_{k+2}, \dots, w_{j+1}\big) \equiv 0. \tag{2.21}
$$

Consequently, all the boundary terms in (2.20) *vanish:*

$$
\begin{aligned}
\mathrm{B}\big(\Lambda; w, \mathrm{D}[w,w]\big) + \mathrm{B}\big(\Lambda; \mathrm{D}[w,w], w\big) + \mathrm{B}\big(\Lambda; \mathrm{D}[w,w], \mathrm{D}[w,w]\big)& \\
+ \sum_{j=2}^{\infty} \sum_{\mathcal{T}\in \mathrm{T}_j} \mathrm{B}\big(\mathcal{T}; w + \mathrm{D}[w,w]\big) \equiv 0.&
\end{aligned} \tag{2.22}
$$

Proof. Write $\mathcal{T}' := \mathcal{T} \overset{\star}{\ominus} \Lambda$, so that $(\mathcal{T}')^0 = \mathcal{T}^0 \setminus \{\star\}$ has cardinality $j-1$. Since $\mathrm{D} = \mathcal{I}(\Lambda, -i\,\mathrm{mul}_{\mathrm{B}}(\Lambda); \cdot)$, the multiplier of the second summand in (2.21) splits as $\mathrm{mul}_{\mathrm{B}}(\mathcal{T}') \cdot \mathrm{mul}_{\mathrm{B}}(\overset{\star}{\Lambda})$, with the auxiliary factor at $\star$ given by $\mathrm{mul}_{\mathrm{B}}(\overset{\star}{\Lambda}) = \mathbb{1}_{A^c_\star} K_\star$. Phase additivity (2.4) forces $\Psi(\mathcal{T}') + \Psi(\overset{\star}{\Lambda}) = \Psi(\mathcal{T})$, so both summands integrate the same exponential over the same domain (after re-indexing the leaves by (2.17)). Adding the multipliers gives the one-line identity

$$
\mathrm{mul}_{\mathrm{B}}(\mathcal{T}) + \mathrm{mul}_{\mathrm{B}}(\mathcal{T}')\, \mathrm{mul}_{\mathrm{B}}(\overset{\star}{\Lambda}) = \big[(-1)^{j+1} + (-1)^j\big] \prod_{\bullet\in\mathcal{T}^0} \mathbb{1}_{A^c_\bullet} K_\bullet = 0, \tag{2.23}
$$

which proves (2.21).

For (2.22), group every boundary contribution in (2.20) whose uncoloured underlying tree is $\mathcal{T}$ (compare with (2.23)); the cumulative multiplier reads

$$
\widetilde{\mathrm{mul}}_{\mathrm{B}}(\mathcal{T}) = \sum_{P\subseteq\mathcal{T}^{0,\mathfrak{f}}} \mathrm{mul}_{\mathrm{B}}\,\big(\mathcal{T} \overset{P}{\ominus} \Lambda\big) \prod_{\star\in P} \mathrm{mul}_{\mathrm{B}}(\overset{\star}{\Lambda}) = (-1)^{j+1} \Big(\prod_{\bullet\in\mathcal{T}^0} \mathbb{1}_{A^c_\bullet} K_\bullet \Big) \sum_{P\subseteq\mathcal{T}^{0,\mathfrak{f}}} (-1)^{|P|},
$$

and vanishes by $\sum_{k=0}^{n} \binom{n}{k} (-1)^k = 0$ with $n = |\mathcal{T}^{0,\mathfrak{f}}| \ge 1$. Identical bookkeeping is carried out term-by-term in [11, Lemma 2.8]. □

Remark 2.9. The cancellation in Lemma 2.8 relies in a crucial way on the fact that the constraint $A^c_\bullet$ on $(\xi_\bullet, \xi_{\bullet 1}, \xi_{\bullet 2})$ depends only on the local node $\bullet$. By contrast, in the standard normal form approach (e.g. [27, 16]) the resonant set involves all ancestors of

$\bullet$, which entangles the trees with their growth history and obliges one to introduce a j-dependent sequence of small parameters to tame the factorial explosion of ordered trees. We refer the reader to [11] for a fuller discussion.

The five surviving tree shapes. The same regrouping principle that erases boundary contributions also drastically thins out the nonlinear side: outside of five distinguished families, every multiplier vanishes. We list the survivors by the location of their final parents:

- **(I)** $\mathcal{T} = \bigwedge$, carrying the bilinear multiplier mul_1 from (2.25).
- **(II)** $j \geq 2$, $\mathcal{T}^{0,\mathfrak{f}} = \{\star\}$ with $\star \neq \emptyset$, the dispersive piece Ψ_0 sitting at $\star$ under $\mathbb{1}_{A_\star}$.
- **(III)** same tree shape as (II), but with the energy piece $E\Psi_E$ at $\star$ under the complementary indicator $\mathbb{1}_{A_\star^c}$; this family is absent in the KdV setting.
- **(IV)** same tree shape as (II)–(III), with the multiplier seated one level higher, at the parent $\mathbf{p}(\star)$ of the unique final parent.
- **(V)** $j \geq 2$, $\mathcal{T}^{0,\mathfrak{f}} = \{\star 1, \star 2\}$ (cherry-tipped): both children of $\star$ are final parents.

A common skeleton emerges: every surviving tree is a root attached to a single chain of *strict* parents, each of which has one child in $\mathcal{T}^\infty$ and the other carrying the chain onwards, terminated either by a single final parent (Types II–IV) or by a sibling pair (Type V). The fact that the infinite-tree children carry no constraint beyond $A_\bullet^c$ is what drives the multilinear estimates of Section 3.

Lemma 2.10. *Let $w : \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}$ be smooth and N as in (2.14). Then*

$$\sum_{j=1}^{\infty} \sum_{\mathcal{T} \in \mathrm{T}_j} \mathrm{N}\big(\mathcal{T}; w + \mathrm{D}[w,w]\big) = \sum_{j=1}^{\infty} \sum_{\mathcal{T} \in \mathrm{T}_j} \sum_{\ell=1}^{5} \mathrm{N}_\ell(\mathcal{T}; w), \tag{2.24}$$

where, for $\ell = 1, \ldots, 5$,

$$\mathcal{F}_x\big[\mathrm{N}_\ell(\mathcal{T}; w)\big] = \mathcal{I}\big(\mathcal{T}, \mathrm{mul}_\ell(\mathcal{T}); w\big),$$

with $\mathcal{I}(\mathcal{T}, m; w)$ as in (2.6), and the multipliers given by

$$\mathrm{mul}_1(\bigwedge) = K\big(\mathbb{1}_A \Psi_0(\bigwedge) - E\Psi_E(\bigwedge)\big), \tag{2.25}$$

$$\begin{aligned}
\mathrm{mul}_2(\mathcal{T}) &= (-1)^{j+1} \mathbb{1}_{A_\star} K_\star \Psi_0(\star) \Big(\prod_{\bullet \in \mathcal{T}^0 \setminus \{\star\}} K_\bullet \mathbb{1}_{A_\bullet^c} \Big) \quad \text{if } \mathcal{T}^{0,\mathfrak{f}} = \{\star\},\ \star \neq \emptyset, \\
\mathrm{mul}_3(\mathcal{T}) &= (-1)^{j} E \mathbb{1}_{A_\star^c} K_\star \Psi_E(\star) \Big(\prod_{\bullet \in \mathcal{T}^0 \setminus \{\star\}} K_\bullet \mathbb{1}_{A_\bullet^c} \Big) \quad \text{if } \mathcal{T}^{0,\mathfrak{f}} = \{\star\},\ \star \neq \emptyset, \\
\mathrm{mul}_4(\mathcal{T}) &= (-1)^{j} K_{\mathbf{p}(\star)} \Psi(\mathbf{p}(\star)) \Big(\prod_{\bullet \in \mathcal{T}^0 \setminus \{\mathbf{p}(\star)\}} K_\bullet \mathbb{1}_{A_\bullet^c} \Big) \quad \text{if } \mathcal{T}^{0,\mathfrak{f}} = \{\star\},\ \star \neq \emptyset, \\
\mathrm{mul}_5(\mathcal{T}) &= (-1)^{j+1} K_\star \Psi(\star) \Big(\prod_{\bullet \in \mathcal{T}^0 \setminus \{\star\}} K_\bullet \mathbb{1}_{A_\bullet^c} \Big), \quad \text{if } \mathcal{T}^{0,\mathfrak{f}} = \{\star 1, \star 2\}, \\
\mathrm{mul}_\ell(\mathcal{T}) &= 0, \quad \ell = 2,3,4,5, \qquad \text{otherwise},
\end{aligned} \tag{2.26}$$

with $\mathcal{T} \in \mathrm{T}_j$, $j \geq 1$, and K and Ψ as in (2.5) *and* (2.3)*, respectively. We refer to the contributions $\mathrm{N}_\ell(\mathcal{T})$ as terms of Type* I, II, III, IV, V*, respectively.*

Proof. The argument follows the inclusion-exclusion strategy of [11, Lemma 2.10, Steps 1–2] verbatim, except for one new ingredient: the bookkeeping triggered by the energy phase $E\Psi_E$, which we record below.

Homogeneity 2. On the l.h.s. of (2.24), only $\mathrm{N}(\bigwedge; w, w)$ contributes at homogeneity 2. Hence $\mathrm{N}_1(\bigwedge; w) = \mathrm{N}(\bigwedge; w, w)$ and $\mathrm{mul}_1(\bigwedge) = \mathrm{mul}(\bigwedge) = K(\mathbb{1}_A \Psi_0(\bigwedge) - E\Psi_E(\bigwedge))$, matching (2.25).

Higher homogeneity. Now fix $j \geq 2$ and $\mathcal{T} \in \mathrm{T}_j$. Following [11, Lemma 2.10], collect all l.h.s. contributions whose underlying uncoloured tree (after expanding each $\mathrm{D}[w,w]$) is $\mathcal{T}$. Indexing such expansions by $C \subseteq \mathcal{T}^{0,\mathfrak{f}}$, the aggregate multiplier is

$$\widetilde{\mathrm{mul}}(\mathcal{T}) = \sum_{C \subseteq \mathcal{T}^{0,\mathfrak{f}}} \mathrm{mul}\,\big(\mathcal{T} \overset{C}{\ominus} \wedge\big) \prod_{\star \in C} \mathrm{mul}_{\mathrm{B}}\big(\overset{\star}{\wedge}\big).$$

The splitting $\Psi(\star) = \Psi_0(\star) + E\Psi_E(\star)$ in the definition of mul (see (2.15)) together with $\mathrm{mul}_{\mathrm{B}}(\overset{\star}{\wedge}) = \mathbb{1}_{A_\star^c} K_\star$ and $(\mathcal{T} \overset{C}{\ominus} \wedge)^0 = \mathcal{T}^0 \setminus C$ decomposes

$$\widetilde{\mathrm{mul}}(\mathcal{T}) = \widetilde{\mathrm{mul}}_{\mathfrak{g},0}(\mathcal{T}) + \widetilde{\mathrm{mul}}_{\mathfrak{g},E}(\mathcal{T}) + \widetilde{\mathrm{mul}}_{\mathfrak{b}}(\mathcal{T}),$$

where $\widetilde{\mathrm{mul}}_{\mathfrak{g},0}$ collects the $\mathbb{1}_{A_\star}\Psi_0(\star)$-terms, $\widetilde{\mathrm{mul}}_{\mathfrak{g},E}$ collects the $\mathbb{1}_{A_\star^c}E\Psi_E(\star)$-terms (*new with respect to KdV*), and $\widetilde{\mathrm{mul}}_{\mathfrak{b}}$ collects the $\Psi(\star)$-terms over strict parents $\star \in \mathcal{T}^0 \setminus \mathcal{T}^{0,\mathfrak{f}}$. Explicitly, the $\mathfrak{g}_E$-piece reads

$$\widetilde{\mathrm{mul}}_{\mathfrak{g},E}(\mathcal{T}) = (-1)^j E\Big(\prod_{\bullet \in \mathcal{T}^0} K_\bullet\Big) \sum_{C \subseteq \mathcal{T}^{0,\mathfrak{f}}} (-1)^{|C|} \sum_{\star \in (\mathcal{T} \overset{C}{\ominus} \wedge)^{0,\mathfrak{f}}} \mathbb{1}_{A_\star^c} \Psi_E(\star) \prod_{\bullet \in \mathcal{T}^0 \setminus \{\star\}} \mathbb{1}_{A_\bullet^c};$$

the expressions for $\widetilde{\mathrm{mul}}_{\mathfrak{g},0}$ and $\widetilde{\mathrm{mul}}_{\mathfrak{b}}$ differ only by the sign, the energy prefactor, and the range of the inner sum.

Step 1 ($\mathfrak{b}$- and $\mathfrak{g}_0$-pieces). The simplifications of $\widetilde{\mathrm{mul}}_{\mathfrak{b}}(\mathcal{T})$ and $\widetilde{\mathrm{mul}}_{\mathfrak{g},0}(\mathcal{T})$ are imported, line for line, from [11, Lemma 2.10, Steps 1–2]: collapsing the alternating sums via the case analysis recorded there gives

$$\widetilde{\mathrm{mul}}_{\mathfrak{b}}(\mathcal{T}) = (-1)^j \Big(\prod_{\bullet \in \mathcal{T}^0} K_\bullet \mathbb{1}_{A_\bullet^c}\Big) \sum_{\star \in \mathcal{T}^0 \setminus \mathcal{T}^{0,\mathfrak{f}}} \Psi(\star)\Big(\mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1\}} + \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 2\}} - \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1,\star 2\}}\Big), \tag{2.27}$$

$$\begin{aligned}\widetilde{\mathrm{mul}}_{\mathfrak{g},0}(\mathcal{T}) = (-1)^{j+1} \sum_{\star \in \mathcal{T}^0} \mathbb{1}_{A_\star} K_\star \Psi_0(\star) \Big(\prod_{\bullet \in \mathcal{T}^0 \setminus \{\star\}} K_\bullet \mathbb{1}_{A_\bullet^c}\Big) \\ \times \Big(\mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star\}} - \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1\}} - \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 2\}} + \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1,\star 2\}}\Big).\end{aligned} \tag{2.28}$$

Step 2 ($\mathfrak{g}_E$-piece, new with respect to KdV). Fix $\star \in \mathcal{T}^0$. Compared to $\widetilde{\mathrm{mul}}_{\mathfrak{g},0}$, the indicator $\mathbb{1}_{A_\star}$ becomes $\mathbb{1}_{A_\star^c}$ and the factor Ψ_0 becomes $E\Psi_E$, while the index set of the inner alternating sum stays put, namely

$$\big\{C \subseteq \mathcal{T}^{0,\mathfrak{f}} \,:\, \star \in (\mathcal{T} \overset{C}{\ominus} \wedge)^{0,\mathfrak{f}}\big\}.$$

Three cases arise.

<u>Case 1: $\star \in \mathcal{T}^{0,\mathfrak{f}}$.</u> The membership $\star \in (\mathcal{T} \overset{C}{\ominus} \wedge)^{0,\mathfrak{f}}$ holds iff $\star \notin C$, so

$$\sum_{\substack{C \subseteq \mathcal{T}^{0,\mathfrak{f}} \\ \star \in (\mathcal{T} \overset{C}{\ominus} \wedge)^{0,\mathfrak{f}}}} (-1)^{|C|} = \sum_{C \subseteq \mathcal{T}^{0,\mathfrak{f}} \setminus \{\star\}} (-1)^{|C|} = \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star\}}.$$

<u>Case 2: $(\star 1, \star 2) \in \mathcal{T}^{0,\mathfrak{f}} \times \mathcal{T}^{\infty}$ (or the symmetric case).</u> Here $\star \in (\mathcal{T} \overset{C}{\ominus} \wedge)^{0,\mathfrak{f}}$ requires $\star 1 \in C$, so the alternating sum collapses to $-\mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1\}}$ (resp. $-\mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 2\}}$).

<u>Case 3: $\star 1, \star 2 \in \mathcal{T}^{0,\mathfrak{f}}$.</u> Now $\star \in (\mathcal{T} \overset{C}{\ominus} \wedge)^{0,\mathfrak{f}}$ requires $\{\star 1, \star 2\} \subseteq C$, yielding $\mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1,\star 2\}}$.

Assembling the three cases gives

$$\widetilde{\mathrm{mul}}_{\mathfrak{g},E}(\mathcal{T}) = (-1)^j E \sum_{\star\in\mathcal{T}^0} \mathbb{1}_{A^c_\star} K_\star \Psi_E(\star)\Big(\prod_{\bullet\in\mathcal{T}^0\setminus\{\star\}} K_\bullet \mathbb{1}_{A^c_\bullet}\Big)$$
$$\times \Big(\mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star\}} - \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1\}} - \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 2\}} + \mathbb{1}_{\mathcal{T}^{0,\mathfrak{f}}=\{\star 1,\star 2\}}\Big). \tag{2.29}$$

At the formal level, (2.29) is obtained from (2.28) via the substitution $(\mathbb{1}_{A_\star}, \Psi_0, (-1)^{j+1}) \mapsto (\mathbb{1}_{A^c_\star}, E\Psi_E, (-1)^j)$, which is the algebraic shadow of $\Psi = \Psi_0 + E\Psi_E$ and is the only ingredient that distinguishes the NV bookkeeping from [11, Lemma 2.10].

Step 3 (combination of the three pieces). Adding (2.27), (2.28), and (2.29) and splitting according to the shape of $\mathcal{T}^{0,\mathfrak{f}}$:

- if $\mathcal{T}^{0,\mathfrak{f}} = \{\star\}$ with $\star \neq \emptyset$, only the indices $\star$ and $\mathbf{p}(\star)$ produce nonzero contributions. At $\star$, $\widetilde{\mathrm{mul}}_{\mathfrak{g},0}$ and $\widetilde{\mathrm{mul}}_{\mathfrak{g},E}$ combine to $\mathrm{mul}_2 + \mathrm{mul}_3$ (Types II and III). At $\mathbf{p}(\star)$, only $\widetilde{\mathrm{mul}}_{\mathfrak{b}}$ contributes, giving mul_4 (Type IV).
- if $\mathcal{T}^{0,\mathfrak{f}} = \{\star 1, \star 2\}$, only the index $\star$ survives; the three pieces combine to mul_5 (Type V).
- otherwise, every alternating sum in (2.27)–(2.29) vanishes, so $\widetilde{\mathrm{mul}}(\mathcal{T}) = 0$.

Consequently, for every $\mathcal{T} \in \mathrm{T}_j$ with $j \geq 2$ the aggregate contribution carried by $\mathcal{T}$ equals $\sum_{\ell=2}^5 \mathcal{I}(\mathcal{T}, \mathrm{mul}_\ell(\mathcal{T}); w)$. Summing over $j \geq 1$ and $\mathcal{T} \in \mathrm{T}_j$ delivers (2.24). □

Remark 2.11. At fixed $j \geq 2$, Types II–III contribute 2^{j-1} trees of size j, and Type IV contributes 2^{j-1} trees of size $j+2$; this is dramatically smaller than $|\mathrm{T}_j| \sim 4^j/j^{3/2}$. The decisive feature of Lemma 2.10, however, is not the cardinality but the rigid shape of the survivors, which forces a very restricted family of frequency interactions and underpins the frequency-restricted estimates in Section 3.

The modified normal form equation and its gauged version. Substituting Lemmas 2.8 and 2.10 into (2.20) produces the modified normal form equation

$$\partial_t w = \sum_{j=1}^\infty \sum_{\mathcal{T}\in\mathrm{T}_j} \sum_{\ell=1}^5 \mathrm{N}_\ell(\mathcal{T}; w).$$

Setting

$$z(t) := U(-t)w(t),$$

with $U(t)$ the linear Novikov–Veselov propagator, a direct calculation shows that z obeys the *gauged Novikov–Veselov equation*

$$\partial_t z + (\partial^3 + \overline{\partial}^3)z + E(\overline{\partial}^{-1}\partial^2 + \partial^{-1}\overline{\partial}^2)z = \sum_{j=1}^\infty \sum_{\mathcal{T}\in\mathrm{T}_j} \sum_{\ell=1}^5 U(t)\mathrm{N}_\ell(\mathcal{T}; U(-t)z) \tag{$\mathcal{G}_\delta$-NV}$$
$$=: \sum_{j=1}^\infty \sum_{\mathcal{T}\in\mathrm{T}_j} \sum_{\ell=1}^5 \mathcal{N}_\ell(\mathcal{T}; z).$$

Equation ($\mathcal{G}_\delta$-NV) is the focus of the next two sections: Section 3 establishes multilinear $X^{s,b}$-bounds for each $\mathcal{N}_\ell(\mathcal{T}; z)$ that are summable in j thanks to the rigid shape of the surviving trees, and Section 4 deduces local well-posedness of ($\mathcal{G}_\delta$-NV) for $H^s(\mathbb{R}^2)$ data throughout the sharp range, before transferring the result back to (NV) via the gauge map.

3. Functional setting and multilinear estimates

Following the conventions of [7, 8, 42], the Fourier restriction spaces tailored to the linearised flow are defined as follows. For exponents $s, b \in \mathbb{R}$, set

$$X^{s,b} = \left\{ u \in \mathcal{S}'(\mathbb{R} \times \mathbb{R}^2) : \langle \tau - \phi(\xi) \rangle^b \langle \xi \rangle^s \widehat{u} \in L^2_{\tau,\xi}(\mathbb{R} \times \mathbb{R}^2) \right\},$$

with the obvious norm. Its restriction to a time slab is, as usual,

$$X^{s,b}_T = \left\{ u \in \mathcal{D}'((0,T) \times \mathbb{R}^2) : u \equiv v\big|_{(0,T)} \text{ for some } v \in X^{s,b} \right\}, \quad T > 0.$$

Road map of the section. Local well-posedness of the gauged Novikov–Veselov equation ($\mathcal{G}_\delta$-NV) hinges on a single multilinear bound, Proposition 3.1 below, in which the implicit constants are tracked uniformly across the infinite family of trees $\mathcal{T} \in \mathrm{T}_j$. Our route to that bound is structural: each tree contribution is funnelled into one of three off-the-shelf frequency-restricted estimates from [13, 14] (Lemmas 3.3, 3.4, 3.5 below), which translate an L^2-integration over the level set $\{|\Phi - \alpha| < M\}$ into a positive power of M. The geometric mechanism underlying every such translation is the NV resonance lemma 3.6: away from a thin stationary stratum, the phase $\Theta = \phi(\xi) - \sum \phi(\xi_j)$ is non-stationary in ξ_1, and on the stationary stratum it admits a uniform Morse normal form with parameters. The proof of Proposition 3.1 is then organised into seven steps, indexed by the tree type $\ell \in \{1, \dots, 5\}$ and by whether the length j is below or above the threshold dictated by the cluster of the focal node $\star$.

Proposition 3.1. *Let $s > -1$, $E \in \mathbb{C}$ with $|E| < 1$, $0 < \delta \ll 1$ and $T > 0$. Given $j \in \mathbb{N}$, $\mathcal{T} \in \mathrm{T}_j$, and $\mathcal{N}_\ell(\mathcal{T})$, $\ell = 1, \dots, 5$, there exists $b > \frac{1}{2}$ such that*

$$\left\| \int_0^t U(t-t') \mathcal{N}_\ell(\mathcal{T}; z_1, \dots, z_{j+1})(t') dt' \right\|_{X^{s,b}_T} \le T^{0+} C^j \delta^{(2s+2)(j-3)-1} \prod_{k=1}^{j+1} \|z_k\|_{X^{s,b}_T} \tag{3.1}$$

for some constant $C > 0$ independent of j and $\mathcal{T}$.

Remark 3.2. For $s > -1$ one has $2s + 2 > 0$, and the seven steps below actually produce the sharper bounds δ^{-1} for $j = 1$, $\delta^{(2s+2)(j-2)}$ in Step 2, 1 for $j = 2, 3$ in Steps 3 and 6, and $\delta^{(2s+2)(j-3)}$ for $j \ge 3$ in Steps 4 and 7. Each of these is dominated by $\delta^{(2s+2)(j-3)-1}$, the only step requiring the extra δ^{-1} factor being Step 1 (Type I). The uniform exponent in (3.1) is convenient for the geometric-series argument in Proposition 4.1, whose summability is guaranteed by (4.1).

The route from Proposition 3.1 to frequency-restricted estimates is by now standard; for the convenience of the reader we collect below the precise statements that we shall plug into our case analysis. The bilinear and higher-multilinear versions appear separately, since the second admits a non-trivial splitting of the multiplier.

Lemma 3.3 ([13, Lemma 3.1])**.** *Let $s \in \mathbb{R}$. Suppose that, for all $\alpha \in \mathbb{R}$ and $M > 1$, there exists $0 < \beta < 1$, such that one of the following holds:*

(i) *for all distinct $k, \ell \in \{\emptyset, 1, 2\}$[2] and $1 < M \lesssim |\alpha|$,*

$$\sup_{\xi_k} \int_{\xi = \xi_1 + \xi_2} \frac{|m_1(\bigwedge)|^2 \langle \xi \rangle^{2s}}{\langle \xi_1 \rangle^{2s} \langle \xi_2 \rangle^{2s}} \mathbb{1}_{|\Phi(\bigwedge) - \alpha| < M} d\xi_\ell \lesssim \langle \alpha \rangle^\beta M^\beta;$$

(ii) *or we have that*

$$\sup_{\xi} \int_{\xi = \xi_1 + \xi_2} \frac{|m_1(\bigwedge)|^2 \langle \xi \rangle^{2s}}{\langle \xi_1 \rangle^{2s} \langle \xi_2 \rangle^{2s}} \mathbb{1}_{|\Phi(\bigwedge) - \alpha| < M} d\xi_1 \lesssim \langle \alpha \rangle^\beta M.$$

Then there exists $b > \frac{1}{2}$ such that (3.1) *holds for $\ell = 1$ and $\mathcal{T} = \bigwedge$.*

[2] The empty-set symbol $\emptyset$ here corresponds to the frequency component ξ, without index.

Lemma 3.4 ([14, Lemma 2]). *Let $s \in \mathbb{R}$, $j \geq 1$, $\ell = 2, \dots, 5$, and $\mathcal{T} \in \mathrm{T}^j$. Let $B \subset \{\emptyset, 1, \dots, j+1\}$ be a proper non-empty subset, and suppose that $\mathcal{M}_1, \mathcal{M}_2 : \mathbb{C}^j \to \mathbb{R}^+$ are multipliers satisfying*

$$\frac{|m_\ell(\mathcal{T})|^2 \langle \xi \rangle^{2s}}{\prod_{k=1}^{j} \langle \xi_k \rangle^{2s}} \lesssim \mathcal{M}_1(\vec{\xi})\, \mathcal{M}_2(\vec{\xi}),$$

where $\vec{\xi} = (\xi_1, \dots, \xi_{j+1})$. If there exists $0 < \beta < 1$ such that

$$\sup_{\xi_{k\in B}} \int_{\Gamma(\mathcal{T})} \mathcal{M}_1(\vec{\xi})\, \mathbb{1}_{|\Phi(\mathcal{T}) - \alpha| < M}\, d\xi_{j \notin B} + \sup_{\xi_{j\notin B}} \int_{\Gamma(\mathcal{T})} \mathcal{M}_2(\vec{\xi})\, \mathbb{1}_{|\Phi(\mathcal{T}) - \alpha| < M}\, d\xi_{j \in B} \lesssim C^j M^\beta,$$

for all $M \geq 1$ and $\alpha \in \mathbb{R}$, then there exists $b > \frac{1}{2}$, depending only on β, such that (3.1) *holds.*

The endpoint $\beta = 1$ in the hypotheses of Lemmas 3.3 and 3.4 is not literally allowed, but it can be reached at the price of a small weight on the largest frequency. The following technical statement formalises this trade-off and will be invoked tacitly in every step of the proof.

Lemma 3.5 ([13, Lemma 3.3]). *Let $K : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^+$ be a positive measurable function such that, for some $N \in \mathbb{N}$,*

$$|K(x, y, M)| \lesssim \max\{1, |x|, |y|\}^N, \quad \forall x \in \mathbb{R}^n,\ y \in \mathbb{R}^m,\ M \in \mathbb{R}.$$

Suppose that

$$\sup_y \int K(x, y, M) \max\{1, |x|, |y|\}^{0+} dx \lesssim M, \quad \textit{for all } M > 1.$$

Then, there exists $0 < \beta < 1$ such that

$$\sup_y \int K(x, y, M) dx \lesssim M^\beta, \quad \textit{for all } M > 1.$$

We turn next to the geometric heart of the proof: the behaviour of the resonance function $\Psi(\mathcal{T})$ on the support of the tree integral. The dichotomy below – non-stationary in ξ_1 off a thin set, Morse with parameters on it – is precisely the analytic content driving the optimal Strichartz estimates obtained in [23, 24] for the linear Novikov–Veselov propagator, and reflects the convexity properties of $\phi(\xi) = \operatorname{Re} \xi^3$.

Lemma 3.6. *Fix $|E| < 1$ and $k \geq 2$. Consider the mapping $\Theta : \mathbb{C}^k \to \mathbb{R}$*

$$\Theta(\xi_1, \dots, \xi_k) = \phi(\xi) - \sum_{j=1}^{k} \phi(\xi_j), \quad \xi = \sum_{j=1}^{k} \xi_j.$$

(i) *Over the region*

$$V = \left\{ \vec{\xi} \in \mathbb{C}^k :\ \frac{1}{\delta} < |\xi_1|,\ |\xi_1 \pm \xi| > \delta \max\{|\xi_1|, |\xi|\} \right\},$$

we have

$$|\nabla_{\xi_1} \Theta| \geq 2\delta^2 \max\{|\xi_1|^2, |\xi|^2\}.$$

(ii) *Suppose that*

$$|\xi_1| \geq |\xi_2| \geq |\xi| \geq |\xi_3| \geq \cdots \geq |\xi_k|.$$

There exist $\delta, \delta' > 0$, independent of k, and a change of coordinates

$$\begin{aligned} \varphi : U = \{\vec{\xi} \in \mathbb{C}^k :\ \tfrac{1}{\delta} < |\xi_1|,\ |\xi_1 + \xi| < \delta |\xi_1|\} &\mapsto B_{\delta'}(0) \times \mathbb{C}^{k-1} \\ \vec{\xi} &\mapsto (q_1, \xi_2, \dots, \xi_k) \end{aligned}$$

satisfying

$$|\det D_{\xi_1}\varphi| \sim |\xi_2|^{-2} \tag{3.2}$$

and

$$\Theta(\vec{\xi}) = |\xi_2|^3 \operatorname{Re} q_1^2 - \sum_{j=2}^{k} \phi(\xi_j).$$

Proof. On the region V, a direct computation yields

$$|\nabla_\xi(\phi_0(\xi) - \phi_0(\xi_1))| = 3|\xi^2 - \xi_1^2| \geq 3\delta^2 \max\{|\xi|^2, |\xi_1|^2\}, \quad |\nabla_\xi(\phi_E(\xi) - \phi_E(\xi_1))| \lesssim 1.$$

Combined with the lower bound $|\xi_1| > 1/\delta$, this gives

$$|\nabla_\xi \Theta| \gtrsim 3\delta^2 \max\{|\xi|^2, |\xi_1|^2\} - 1 \gtrsim 2\delta^2 \max\{|\xi|^2, |\xi_1|^2\},$$

proving (i).

We now turn to (ii). The two ingredients driving the argument are (a) the comparability $|\xi_1| \leq 2|\xi_2|$ on U, which makes $|\xi_2|$ the natural unit of length, and (b) the perturbative smallness $|E|/|\xi_2|^2 < \delta^2$ of the energy term. We exploit them through the rescaling

$$p_j = \frac{\xi_j}{|\xi_2|}, \quad j = \emptyset, 1, \ldots, k,$$

which puts Θ into the form

$$\Theta = |\xi_2|^3 \left(\phi_0(p) - \sum_{j=1}^{k} \phi_0(p_j) - \frac{E}{|\xi_2|^2} \left(\phi_E(p) - \sum_{j=1}^{k} \phi_E(p_j) \right) \right). \tag{3.3}$$

A short manipulation, using $c = \sum_{j=2}^{k} p_j = p - p_1$ and the shifted variable $p_1' = c^{\frac{1}{2}}\left(p_1 + \frac{c}{2}\right)$, identifies the principal part:

$$\phi_0(p) - \phi_0(p_1) - \phi_0(c) = 3\operatorname{Re}(p_1')^2.$$

The pieces $\phi_0(c)$ and $\sum_{j\geq 2} \phi_0(p_j)$ depend only on $p_2, \ldots, p_k$, so they will eventually be reabsorbed into $\sum \phi(\xi_j)$ on the right of (3.3) once we revert to ξ-variables. The remaining variable c ranges over the compact annulus $C := \{c \in \mathbb{C} : 1/3 < |c| < 3\}$, and the bound $|p_1'| < 4\delta$ holds throughout U.

It is convenient to package the energy contribution as a parameter:

$$\varepsilon = \frac{E}{|\xi_2|^2}, \qquad |\varepsilon| < \delta^2.$$

With this choice, the function

$$F(p_1', c, \varepsilon) = 3\operatorname{Re}(p_1')^2 - \varepsilon\big(\phi_E(p_1 + c) - \phi_E(p_1)\big)$$

becomes the object on which the Morse normal form is to be applied. Pick a reference point $c_0 \in C$. At $\varepsilon = 0$, the map $p_1' \mapsto F(p_1', c_0, 0)$ has an isolated critical point at $p_1' = 0$ whose real Hessian has signature $(1,1)$. Invoking the Morse lemma with parameters, in the form available in Hörmander, *The Analysis of Linear Partial Differential Operators*, vol. I, Lemma C.6.1 (or Duistermaat, *Fourier Integral Operators*, Section 3.4), we obtain radii $\eta(c_0), \eta'(c_0) > 0$ and a smooth family $\varphi_{c,\varepsilon} : B_{\eta(c_0)}(0) \to B_{\eta'(c_0)}(0)$ of diffeomorphisms $\varphi_{c,\varepsilon}(p_1') = q_1$ such that, for $|c - c_0| + |\varepsilon| < \eta(c_0)$,

$$F(p_1', c, \varepsilon) = \operatorname{Re} q_1^2, \qquad |\det D_{p_1'}\varphi_{c,\varepsilon}| \sim 1.$$

The next step is to make the radii independent of c via compactness. The family $\{|c - c_0| < \eta(c_0)\}_{c_0 \in C}$ covers the compact annulus C, so we can extract a finite subcover indexed by $c_0^{(1)}, \ldots, c_0^{(N)}$ and define

$$\delta = \min_{1\leq i\leq N} \eta(c_0^{(i)}), \qquad \delta' = \min_{1\leq i\leq N} \eta'(c_0^{(i)}).$$

After shrinking δ further to ensure $\delta^2 < \delta$, the constants depend on E (through the upper bound $|E| < 1$) but neither on the residual variables $p_2, \dots, p_k$ nor on the height k of the tree.

Going back to the original coordinates,

$$\Theta = |\xi_2|^3 F\left(p_1', c, \varepsilon\right) - \sum_{j=2}^{k} \phi(\xi_j) = |\xi_2|^3 \operatorname{Re} q_1^2 - \sum_{j=2}^{k} \phi(\xi_j).$$

The Jacobian (3.2) of the resulting map $\varphi : \xi_1 \mapsto q_1$ follows from chaining the rescaling $\xi_1 = |\xi_2| p_1$ with the unit-Jacobian estimate $|\det D_{p_1'} \varphi_{c,\varepsilon}| \sim 1$. $\square$

Remark 3.7. By trivial permutations, the conclusions of Lemma 3.6 also hold for other pairings of frequencies.

For convenience, we record the pointwise estimates for the multipliers $\mathrm{mul}_\ell(\mathcal{T})$, $\ell = 1, \dots, 5$, of Lemma 2.10 in the following lemma, whose proof is immediate.

Lemma 3.8. *Given $\xi_1, \xi_2 \in \mathbb{C}$ and $\xi = \xi_1 + \xi_2$,*

$$|\Psi_0(\bigwedge)| \lesssim |\xi\xi_1\xi_2|, \quad |\Psi_E(\bigwedge)| \gtrsim \min_{j=\emptyset,1,2} |\xi_j|, \quad |K| \lesssim \frac{1}{|\xi_1||\xi_2|}.$$

If $|\xi| \gtrsim 1$, then $|\Psi(\bigwedge)| \lesssim |\xi\xi_1\xi_2|$.

Proof of Proposition 3.1. The seven steps that follow are arranged primarily by tree type ℓ and, within each type, by whether the length j falls below or above the smallest value for which the cluster of $\star$ is entirely interior to $\mathcal{T}$.

Step 1. Type I, $j = 1$, $\mathcal{T} = \bigwedge$: workhorse Lemma 3.3-(ii). The bilinear estimate at $j = 1$ is the only one not subsumed by Lemma 3.4; we split the multiplier $\mathrm{mul}_1(\bigwedge)$ into two pieces and feed each into Lemma 3.3-(ii) with $\beta = 1$, the residual $\beta = 1$ being relaxed to $\beta < 1$ at the end via Lemma 3.5. We have

$$|\,\mathrm{mul}_1(\bigwedge)| \lesssim |\xi| \mathbb{1}_A + K|\Psi_E(\bigwedge)| \mathbb{1}_{A^c} \lesssim |\xi| \mathbb{1}_A + \frac{\mathbb{1}_{A^c}}{\max_{\bullet \in \{\emptyset,1,2\}} |\xi_\bullet|}.$$

Without loss of generality, $|\xi_1| \gtrsim |\xi_2|$. For the first term, $|\xi_2| \lesssim 1/\delta$ on A and $|\partial_{\xi_2} \Psi(\bigwedge)| \gtrsim |\xi|^2$, so that

$$\sup_\xi \int_A \frac{|\xi|^2 \langle \xi \rangle^{2s}}{\langle \xi_1 \rangle^{2s} \langle \xi_2 \rangle^{2s}} \mathbb{1}_{|\Psi(\bigwedge) - \alpha| < M} d\xi_2 \lesssim \sup_\xi \int_{|\zeta_2| < 1/\delta} \mathbb{1}_{|\phi - \alpha| < M}\, d\phi\, d\zeta_2 \lesssim \delta^{-1} M.$$

Lemma 3.5 converts the $\delta^{-1}M$ on the right-hand side into $\delta^{-1}M^\beta$ for some $\beta < 1$ (the δ^{-1} is harmless as it carries no M-dependence), so Lemma 3.3-(ii) applies.

For the second term, we may assume that $\xi \not\approx \pm\xi_2$ (otherwise $\xi_1 \not\approx \pm\xi_2$ and we swap the roles of ξ and ξ_1), so that $|\partial_{\xi_2} \Psi(\bigwedge)| \gtrsim |\xi|^2$. Then

$$\sup_\xi \int \frac{\langle \xi \rangle^{2s}}{\langle \xi_1 \rangle^{2s+2-} \langle \xi_2 \rangle^{2s}} \mathbb{1}_{|\Psi(\bigwedge) - \alpha| < M} d\xi_2 \lesssim \sup_\xi \int \frac{1}{|\zeta_2|^{4+2s-}} \mathbb{1}_{|\phi - \alpha| < M}\, d\phi\, d\zeta_2 \lesssim M,$$

and Lemmas 3.3-(ii) and 3.5 apply once more.

The output is the bound δ^{-1}, weaker than the uniform $\delta^{(2s+2)(j-3)-1} = \delta^{-4s-5}$ (cf. Remark 3.2), but with the explicit δ^{-1} that controls the geometric series in j.

For Types II–V (i.e. $\ell = 2, \dots, 5$) the workhorse switches from Lemma 3.3 to Lemma 3.4, whose flexibility in choosing the splitting set B is essential. Lemma 3.5 is then applied at the very end of each step.

Step 2. Type II, $\ell = 2$, $j \geq 2$: workhorse Lemma 3.4 with $B = \{\star 2\}$. The Type II multiplier carries an extra inverse weight at the parent leaf of $\star$. Choosing $B = \{\star 2\}$ splits the integrations so that resonance is traded only against $\xi_{\star 2}$ (yielding M), while each of the $j-2$ remaining internal nodes contributes a factor δ^{2s+2} via the small-frequency restriction $|\xi_{\bullet 2}| \lesssim 1/\delta$; the cumulative effect is $\delta^{(2s+2)(j-2)}M$. We have, for $\mathcal{T} \in \mathrm{T}_j$, $j \geq 2$,

$$|\operatorname{mul}_2(\mathcal{T})| \lesssim \frac{\langle \xi \rangle^s}{\langle \xi_{\star 1} \rangle^s \langle \xi_{\star 2} \rangle^s \langle \xi_{\mathbf{s}(\star)} \rangle^{s+1}} \prod_{\substack{\bullet \in \mathcal{T}^0 \\ \bullet \neq \star, \mathbf{p}(\star)}} \frac{1}{\langle \xi_{\bullet 1} \rangle \langle \xi_{\bullet 2} \rangle^{s+1}}.$$

On $A_\star$, we have $|\xi_{\star 2}| \lesssim 1/\delta$. The worst-case scenario occurs when $\xi_{\star 1}$ corresponds to the largest frequency (the specific tree structure plays no further role).[3] For ξ and $\xi_{\bullet 2}$, $\bullet \in \mathcal{T}^0 \setminus \{\star 1\}$, fixed,

$$\int |\xi_{\star 1}|^2 \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M}\, d\xi_{\star 2} \lesssim \int \mathbb{1}_{|\phi - \alpha| < M}\, d\phi\, d\zeta_{\star 2} \lesssim M.$$

On the other hand, for $\xi_{\star 1}, \xi_{\star 2}$ fixed, the resonance restriction is not used:

$$\int \frac{\langle \xi \rangle^{2s}}{\langle \xi_{\star 1} \rangle^{2s+2-} \langle \xi_{\mathbf{s}(\star)} \rangle^{2s+2}} \cdot \prod_{\substack{\bullet \in \mathcal{T}^0 \\ \bullet \neq \star, \mathbf{p}(\star)}} \frac{d\xi_{\bullet 2}}{\langle \xi_{\bullet 1} \rangle^2 \langle \xi_{\bullet 2} \rangle^{2s+2}} \cdot \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M}\, d\xi$$

$$\lesssim \int \left(\int \frac{\langle \xi \rangle^{2s}\, d\xi}{\langle \xi_{\star 1} \rangle^{2s+2-} \langle \xi_{\mathbf{s}(\star)} \rangle^{2s+2}} \right) \prod_{\substack{\bullet \in \mathcal{T}^0 \\ \bullet \neq \star, \mathbf{p}(\star)}} \frac{d\xi_{\bullet 2}}{\langle \xi_{\bullet 1} \rangle^2 \langle \xi_{\bullet 2} \rangle^{2s+2}} \lesssim \delta^{(2s+2)(j-2)}.$$

Plugging both bounds into Lemma 3.4 and then upgrading via Lemma 3.5 delivers (3.1).

Step 3. Type IV, $\ell = 4$, $j = 2$, $\mathcal{T} =$: workhorse Lemma 3.4 with $B = \{11\}$, supported by Lemma 3.6. With $j = 2$ the only tree is $\mathcal{T} =$. Both quantities (3.5)–(3.6) below are estimated by integrating along the resonance level set; the genuinely delicate situation is the stationary one (Case B2 below), which is resolved by invoking the Morse change of coordinates of Lemma 3.6-(ii). With $\mathcal{T} =$,

$$|\operatorname{mul}_4(\mathcal{T})| \lesssim \frac{\langle \xi \rangle^{s+1}}{\langle \xi_{11} \rangle^{s+1} \langle \xi_{12} \rangle^{s+1} \langle \xi_2 \rangle^s}. \tag{3.4}$$

The worst-case scenario corresponds to $|\xi_2| \geq |\xi| \geq |\xi_{11}| \geq |\xi_{12}|$. The proof proceeds by bounding

$$\sup_{\xi, \xi_{12}, \alpha} \int |\xi|^{1-} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M}\, d\xi_{11} \tag{3.5}$$

and

$$\sup_{\xi_2, \xi_{11}, \alpha} \int \frac{|\xi|^{1+}}{\langle \xi_{12} \rangle^{4s+4}} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M}\, d\xi_{12}. \tag{3.6}$$

Case A. $\xi \not\simeq \pm\xi_{12}$ and $\xi_2 \not\simeq \pm\xi_{11}$.
For ξ, ξ_{12} fixed,

$$(3.5) \lesssim \int \frac{1}{|\xi_2|^{1+}} \mathbb{1}_{|\phi - \alpha| < M}\, d\phi\, d\zeta_{11} \lesssim M.$$

For ξ_2, ξ_{11} fixed,

$$(3.6) \lesssim \int \frac{1}{\langle \xi_{12} \rangle^{4s+4} |\xi|^{1-}} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M}\, d\phi\, d\zeta_{12} \lesssim M.$$

Case B. $\xi \simeq \pm\xi_{12}$ or $\xi_2 \simeq \pm\xi_{11}$. Then all four frequencies are comparable, so that

$$|\operatorname{mul}_4(\mathcal{T})| \lesssim |\xi|^{-1-2s}.$$

Case B1. Up to permutations, the conditions of Case A are met; it suffices to repeat the arguments therein.

[3]We can suppose this because the bound is symmetric in the leaves once we have isolated $\star$.

Case B2. Three of the frequencies are equal up to a sign, say $\xi_2 \simeq \xi_{11} \simeq \pm\xi_{12}$. If the sign is positive, then

$$\xi_2 \simeq \xi_{11} \simeq \xi_{12} \simeq \tfrac{\xi}{3}.$$

For ξ_2, ξ_{11} fixed we proceed as in Case A (the non-stationarity condition still holds). For ξ, ξ_{12} fixed, we apply Lemma 3.6-(ii) (with ξ_1 replaced by ξ_{11}, ξ replaced by $\xi - \xi_{12}$, and parameters rescaled by $|\xi_2|$) to obtain

$$(3.5) \lesssim \int_{|q_1|\ll 1} |\xi_2|^{3-} \mathbb{1}_{||\xi|^3 \operatorname{Re} q_1^2 - \widetilde{\alpha}|<M} \, dq_1 \lesssim M^{1-}.$$

Finally, if $\xi_2 \simeq \xi_{11} \simeq -\xi_{12}$, then $\xi \simeq -\xi_{12}$. In this case, both (3.5) and (3.6) are handled by Lemma 3.6-(ii) as above.

The case $j \geq 3$, handled next, differs only in that the $j-3$ internal nodes outside the cluster of $\star$ contribute free δ^{2s+2} factors via Ψ_E.

Step 4. Type IV, $\ell = 4$, $j \geq 3$: workhorse Lemma 3.4, with the $j-3$ free integrations accumulating $\delta^{(2s+2)(j-3)}$. At lengths $j \geq 3$ the tree has $j-3$ additional internal nodes outside the cluster of $\star$. Each contributes an unrestricted integration on $|\xi_{\bullet 2}| \lesssim 1/\delta$, and the budget $\delta^{(2s+2)(j-3)}$ in the final bound comes entirely from these. The dichotomy of Step 3 – non-stationary (Case A) versus stationary (Case B) – is reproduced verbatim at the level of (3.8)–(3.9). Set $\circ = \mathbf{g}(\star)$. Then

$$|\operatorname{mul}_4(\mathcal{T})| \lesssim \frac{\langle \xi \rangle^s}{\langle \xi_{\circ 111} \rangle^{s+1} \langle \xi_{\circ 112} \rangle^{s+1} \langle \xi_{\circ 12} \rangle^{s} \langle \xi_{\circ 2} \rangle^{s+1}} \prod_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \circ} \frac{1}{\langle \xi_\bullet \rangle^{s+1} \langle \xi_{\mathbf{s}(\bullet)} \rangle}. \tag{3.7}$$

We focus on the worst-case scenario $|\xi_{\circ 12}| \geq |\xi_{\circ 2}| \geq |\xi_{\circ 111}| \geq |\xi_{\circ 112}| \geq |\xi|$. Consider

$$\sup_{\xi_{\circ 111}, \xi_{\circ 12}} \int \frac{\langle \xi \rangle^{2s}}{|\xi_{\circ 12}|^{1+2s}} \left(\prod_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \circ} \frac{d\xi_\bullet}{\langle \xi_\bullet \rangle^{2s+2} \langle \xi_{\mathbf{s}(\bullet)} \rangle^2} \right) \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M} \, d\xi \, d\xi_{\circ 112} \tag{3.8}$$

and

$$\sup_{\xi, \xi_{\circ 2}, \xi_{\circ 112}, \xi_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \circ}} \int \frac{|\xi_{\circ 12}|^{1+}}{\langle \xi_{\circ 111} \rangle^{2s+2} \langle \xi_{\circ 112} \rangle^{2s+2} \langle \xi_{\circ 2} \rangle^{2s+2}} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M} \, d\xi_{\circ 111}. \tag{3.9}$$

Case A. $\xi_{\circ 12} \not\simeq \pm\xi_{\circ 111}$ and $\xi_{\circ 2} \not\simeq \pm\xi_{\circ 112}$.
For $\xi_{\circ 12}, \xi_{\circ 111}$ fixed,

$$(3.8) \lesssim \int \frac{\langle \xi \rangle^{2s}}{|\xi_{\circ 12}|^{3+2s}} \left(\prod_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \circ} \frac{d\xi_\bullet}{\langle \xi_\bullet \rangle^{2s+2} \langle \xi_{\mathbf{s}(\bullet)} \rangle^2} \right) \mathbb{1}_{|\phi - \alpha| < M} \, d\xi \, d\zeta_{\circ 112} \lesssim \delta^{(2s+2)(j-3)} M,$$

where the $j-3$ factors of δ^{2s+2} originate from the $|\xi_{\bullet 2}| \lesssim 1/\delta$ constraints associated with each of the $j-3$ internal vertices not lying above $\circ$. Similarly, for $\xi, \xi_{\circ 2}, \xi_{\circ 112}$ and $\xi_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \circ}$ fixed,

$$(3.9) \lesssim \int \frac{1}{\langle \xi_{\circ 111} \rangle^{2s+2} |\xi_{\circ 2}|^{1-}} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M} \, d\phi \, d\zeta_{\circ 111} \lesssim M.$$

Case B. $\xi_{\circ 12} \simeq \pm\xi_{\circ 111}$ or $\xi_{\circ 2} \simeq \pm\xi_{\circ 112}$.
We proceed as in the corresponding Case B of Step 3, applying if necessary the change of coordinates of Lemma 3.6-(ii); the $\delta^{(2s+2)(j-3)}$ factor again arises from the $j-3$ free integrations.

Step 5. Type III, $\ell = 3$, $j \geq 2$: reduction to Type IV (Steps 3–4). Type III is not analysed on its own: the bound on $\operatorname{mul}_3(\mathcal{T})$ carries an extra inverse weight $\langle \xi_{\mathbf{s}(\star)} \rangle^{-(s+1)}$ relative to

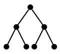
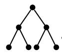
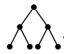
$\mathrm{mul}_4(\mathcal{T})$, hence any frequency-restricted estimate valid for the latter applies a fortiori to the former. Indeed, given $\mathcal{T} \in \mathrm{T}_j$, $j \geq 2$, the estimate $|\Psi_E(\star)| \lesssim |\xi_\star|$ yields

$$|\,\mathrm{mul}_3(\mathcal{T})| \lesssim \frac{\langle \xi \rangle^s}{\langle \xi_{\star 1}\rangle^{s+1}\langle \xi_{\star 2}\rangle^{s+1}\langle \xi_{\mathbf{s}(\star)}\rangle^{s+1}} \prod_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \mathbf{p}(\star)} \frac{1}{\langle \xi_\bullet \rangle \langle \xi_{\mathbf{s}(\bullet)}\rangle^{s+1}}.$$

If $j = 2$, this is a stronger bound than (3.4). If $j \geq 3$, with $\circ = \mathbf{g}(\star)$, this estimate can be rewritten as

$$|\,\mathrm{mul}_3(\mathcal{T})| \lesssim \frac{\langle \xi \rangle^s}{\langle \xi_{\circ 111}\rangle^{s+1}\langle \xi_{\circ 112}\rangle^{s+1}\langle \xi_{\circ 12}\rangle^{s+1}\langle \xi_{\circ 1}\rangle\langle \xi_{\circ 2}\rangle^{s+1}} \prod_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \circ} \frac{1}{\langle \xi_\bullet \rangle \langle \xi_{\mathbf{s}(\bullet)}\rangle^{s+1}},$$

which is smaller than the bound (3.7). In either case the Type III estimate follows from the proof of Steps 3 and 4 verbatim.

Step 6. Type V, $\ell = 5$, $j = 3$, $\mathcal{T} =$ *: workhorse Lemma 3.4, supported by Lemma 3.6.* At $j = 3$, Type V is realised by the single tree $\mathcal{T} =$ , whose multiplier involves four leaves rather than three. The bounds for the two integrals (3.10)–(3.11) extend those of Step 3 by one resonance-free integration, again resolved either by non-stationarity (Case A) or, on the stationary stratum, by Lemma 3.6-(ii) (Case B2). With $\mathcal{T} =$ ,

$$|\,\mathrm{mul}_5(\mathcal{T})| \lesssim \frac{\langle \xi \rangle^{s+1}}{\langle \xi_{11}\rangle^{s+1}\langle \xi_{12}\rangle^{s+1}\langle \xi_{21}\rangle^{s+1}\langle \xi_{22}\rangle^{s+1}}.$$

The worst-case scenario occurs for $|\xi| \geq |\xi_{11}| \geq |\xi_{12}| \geq |\xi_{21}| \geq |\xi_{22}|$, and we consider

$$\sup_{\xi, \xi_{12}, \xi_{22}} \int |\xi|^{1-} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M} \, d\xi_{21} \tag{3.10}$$

and

$$\sup_{\xi_{11}, \xi_{21}} \int \frac{1}{|\xi|^{1-}\langle \xi_{12}\rangle^{2s+2}\langle \xi_{21}\rangle^{2s+2}\langle \xi_{22}\rangle^{2s+2}} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M} \, d\xi_{12} \, d\xi_{22}. \tag{3.11}$$

Case A. $\xi \not\simeq \pm\xi_{12}$ and $\xi_{11} \not\simeq \pm\xi_{21}$.
For ξ, ξ_{12}, ξ_{22} fixed,

$$(3.10) \lesssim \int \frac{1}{|\xi|^{1+}} \mathbb{1}_{|\phi - \alpha| < M} \, d\phi \, d\zeta_{21} \lesssim M. \tag{3.12}$$

For ξ_{11}, ξ_{21} fixed,

$$(3.11) \lesssim \int \frac{1}{|\xi_{11}|^{1-}\langle \zeta_{12}\rangle^{2s+2}\langle \xi_{22}\rangle^{4s+4}} \mathbb{1}_{|\phi - \alpha| < M} \, d\phi \, d\zeta_{21} \, d\xi_{22} \lesssim M. \tag{3.13}$$

Case B. $\xi \simeq \pm\xi_{12}$ or $\xi_{11} \simeq \pm\xi_{21}$. In particular, $|\xi| \sim |\xi_{21}|$. We split further:
Case B1. Up to a permutation of $\xi, \xi_{11}, \xi_{12}, \xi_{21}$, the conditions of Case A are met.
Case B2. The frequencies $\xi_{11}, \xi_{12}, \xi_{21}$ are equal up to a sign. Up to permutation we may assume

$$\xi \not\simeq \xi_{12} \quad \text{and} \quad \xi_{11} \not\simeq -\xi_{21}.$$

For ξ, ξ_{12}, ξ_{22} fixed, either $\xi_{11} \not\simeq -\xi_{21}$ (and we bound (3.10) as in (3.12)) or $\xi_{11} \simeq -\xi_{21}$. In the latter sub-case, Lemma 3.6-(ii) yields

$$(3.10) \lesssim \int |\xi|^{3-} \mathbb{1}_{||\xi|^2 \operatorname{Re} q_1^2 - \widetilde{\alpha}| < M} \, dq_1 \lesssim M^{1-}.$$

For ξ_{11}, ξ_{21} fixed we argue analogously, splitting between the non-stationary case (handled by (3.13)) and the stationary one (handled by Lemma 3.6-(ii)).

In the next step we treat Type V with $j \geq 4$, where the additional internal nodes contribute $j - 3$ free integrations and the new leaf $\xi_{\mathbf{s}(\star)}$ requires an extra resonance-free integration that is absorbed by the $|\xi_{\star 11}|^2$ weight in mul_5.

Step 7. Type V, $\ell = 5$, $j \geq 4$: workhorse Lemma 3.4, with one additional integration absorbed by the $|\xi_{\star 11}|^2$ weight. For lengths $j \geq 4$, Type V exhibits a structural feature that has no counterpart in Step 4: the node $\mathbf{s}(\star)$, which is internal for Type IV trees, becomes a leaf of $\mathcal{T}_5$ and thus contributes a free integration variable. The compensation is built into mul_5 via an extra inverse weight $|\xi_{\star 11}|^{-2}$ absent from mul_4, and below we trace how this weight precisely absorbs the new variable while leaving the $\delta^{(2s+2)(j-3)}$ accounting of Step 4 intact. The multiplier is now

$$|\,\mathrm{mul}_5(\mathcal{T})| \lesssim \frac{\langle \xi \rangle^s}{\langle \xi_{\star 11}\rangle^{s+1}\langle \xi_{\star 12}\rangle^{s+1}\langle \xi_{\star 21}\rangle^{s+1}\langle \xi_{\star 22}\rangle^{s+1}\langle \xi_{\mathbf{s}(\star)}\rangle^{s+1}} \prod_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \mathbf{p}(\star)} \frac{1}{\langle \xi_{\mathbf{s}(\bullet)}\rangle \langle \xi_\bullet \rangle^{s+1}}.$$

In the worst-case scenario, $|\xi_{\star 11}| \geq |\xi_{\star 12}| \geq |\xi_{\star 21}| \geq |\xi_{\star 22}| \geq |\xi_{\mathbf{s}(\star)}| \geq |\xi|$. We consider

$$\sup_{\xi_{\star 11}, \xi_{\star 21}, \xi_{\mathbf{s}(\star)}} \int \frac{\langle \xi \rangle^{2s}}{|\xi_{\star 11}|^{1+2s}} \prod_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \mathbf{p}(\star)} \frac{d\xi_\bullet}{\langle \xi_{\mathbf{s}(\bullet)}\rangle^2 \langle \xi_\bullet \rangle^{2s+2}} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M} \, d\xi \, d\xi_{\star 22} \tag{3.14}$$

and

$$\sup_{\xi, \xi_{\star 12}, \xi_{\star 22}, \xi_{\bullet \in \mathcal{T}^\infty, \bullet \not\geq \mathbf{p}(\star)}} \int \frac{1}{|\xi_{\star 11}|^{1-} \langle \xi_{\star 21}\rangle^{4s+4} \langle \xi_{\mathbf{s}(\star)}\rangle^{4s+4}} \mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M} \, d\xi_{\star 21} \, d\xi_{\mathbf{s}(\star)}. \tag{3.15}$$

The quantity (3.14) is structurally identical to (3.8) of Step 4: the integration $d\xi$ runs over the resonance level set and produces M as in Case A of Step 4, while the $j-3$ free integrations on $|\xi_{\bullet 2}| \lesssim 1/\delta$ associated with the internal nodes not lying above $\mathbf{p}(\star)$ accumulate the factor $\delta^{(2s+2)(j-3)}$.

The quantity (3.15) is the genuine novelty of Step 7. Compared with its Step 4 analogue (3.9), it carries *one additional integration* variable, namely $\xi_{\mathbf{s}(\star)}$ (the parent of the cluster of $\star$, which is a leaf in $\mathcal{T}_5$ but an internal node in $\mathcal{T}_4$). The compensation comes from a single *extra inverse weight* $|\xi_{\star 11}|^{-2}$ that is present in mul_5 but absent in mul_4: using $|\xi_{\star 11}| \geq |\xi_{\mathbf{s}(\star)}|$ in the worst-case ordering, one has

$$\frac{1}{|\xi_{\star 11}|^{1-} \langle \xi_{\star 21}\rangle^{4s+4} \langle \xi_{\mathbf{s}(\star)}\rangle^{4s+4}} \lesssim \frac{1}{|\xi_{\star 11}|^{1-} \langle \xi_{\star 21}\rangle^{4s+4} \langle \xi_{\mathbf{s}(\star)}\rangle^{4s+2} \, |\xi_{\star 11}|^2},$$

so that the integration in $\xi_{\mathbf{s}(\star)}$ converges as $\int \langle \xi_{\mathbf{s}(\star)} \rangle^{-(4s+2)} \, d\xi_{\mathbf{s}(\star)} \lesssim 1$ for $s > -1$, and the surviving $|\xi_{\star 11}|^{-(3-)}$ together with the resonance restriction $\mathbb{1}_{|\Psi(\mathcal{T}) - \alpha| < M}$ on $\xi_{\star 21}$ produces the bound M exactly as in Case A of Step 4. In other words, the extra integration of Step 7 introduces *one extra parent node* ($\mathbf{s}(\star)$) compared with Step 4, and the $|\xi_{\star 11}|^2$ weight in mul_5 supplies precisely the δ^{2s+2} budget required to absorb it without consuming resonance. The bookkeeping of the remaining $j-3$ free integrations is identical to Step 4, and the end result is again $\delta^{(2s+2)(j-3)} M$. An application of Lemma 3.4 combined with Lemma 3.5 closes (3.1).

Assembling Steps 1–7 yields, for every $\mathcal{T} \in \mathrm{T}_j$ and every $\ell \in \{1, \ldots, 5\}$, the estimate (3.1) with the uniform exponent $\delta^{(2s+2)(j-3)-1}$ (cf. Remark 3.2). This is precisely the input absorbed by the contraction argument in Proposition 4.1: under the smallness assumption (4.1), the geometric series $\sum_{j \geq 1} C^j \delta^{(2s+2)(j-3)-1} R^{j+1}$ sums to $O(\delta^{-4s-5} R^2)$, with the dominant contribution coming from the Type-I term. □

4. Proof of the main theorem

In this section, we prove Theorem 1.1. By a suitable rescaling, it suffices to consider the case $|E| < 1$. Throughout this section, we fix $R \gg 1$ and take $\delta > 0$ such that

$$\delta^{2s+2} R \ll 1. \tag{4.1}$$

Proposition 4.1 (Local well-posedness for the gauged NV equation)**.** *Fix $s>-1$. Given $z_0\in H^s(\mathbb{R}^2)$ with $\|z_0\|_{H^s}<R$, there exist $T=T_{\delta,R}>0$ and a unique*

$$z\in X^{s,\frac12+}(0,T)$$

solution of ($\mathcal{G}_\delta$-NV) over $(0,T)$ with initial data z_0.

Proof. The argument is by Banach fixed point in the standard way; we record the main inequality in order to highlight the dependence on δ.

Fix $b=\frac12+$ and define

$$B_{R,T}=\left\{z\in X^{s,b}_T:\|z\|_{X^{s,b}(0,T)}\le 2R\right\},$$

endowed with the induced distance, together with the integral map

$$\Theta[z](t)=U(t)z_0+\int_0^t U(t-t')\sum_{j=1}^\infty\sum_{\mathcal{T}\in\mathrm{T}_j}\sum_{\ell=1}^5\mathcal{N}_\ell(\mathcal{T};z)(t')\,dt'.$$

Applying the multilinear estimates of Proposition 3.1 and recalling that $|\mathrm{T}_j|$ grows only exponentially in j (cf. (2.1)), for any $z\in B_{R,T}$ we obtain

$$\begin{aligned}\|\Theta[z]\|_{X^{s,b}(0,T)}&\le\|z_0\|_{H^s}+T^{0+}\sum_{j=1}^\infty C^j\delta^{(2s+2)(j-3)-1}\|z\|^{j+1}_{X^{s,b}(0,T)}\\&\le R+T^{0+}\delta^{-4s-5}\sum_{j=1}^\infty\left(C\,\delta^{2s+2}\,R\right)^{j-1}\cdot CR^2\\&\le R+T^{0+}C\,\delta^{-4s-5}\,R^2\le 2R,\end{aligned}$$

where the geometric series converges thanks to the smallness condition (4.1) and $T=T_{\delta,R}$ is chosen sufficiently small in the last inequality. Using the multilinear structure of the nonlinear terms, an analogous computation yields

$$\|\Theta[z]-\Theta[z']\|_{X^{s,b}(0,T)}\le T^{0+}C\delta^{-4s-5}R\|z-z'\|_{X^{s,b}(0,T)}\le\tfrac12\|z-z'\|_{X^{s,b}(0,T)}.$$

The result follows from Banach's fixed point theorem. □

Remark 4.2. If $z_0\in H^{s'}(\mathbb{R}^2)$, for some $s'>s$, a standard persistence-of-regularity argument shows that $z\in X^{s',b}(0,T)\hookrightarrow C([0,T];H^s(\mathbb{R}^2))$.

Recall the (formal) definition of the gauge transform,

$$\mathcal{G}_\delta^{-1}[f]=f+\mathcal{F}_z^{-1}\left(\int_{\xi=\xi_1+\xi_2}K\mathbb{1}_{A_\delta^c}(\xi_1,\xi_2)\widehat{f}(\xi_1)\widehat{f}(\xi_2)d\xi_1\right).\tag{4.2}$$

Lemma 4.3. *Consider the map $\mathcal{G}^{-1}$ defined in (4.2).*

(i) *Let $s'\ge s>-1$, $R>0$, $0<\delta\ll R^{-\frac{1}{2s+2}}$, and $B_R^{s',s}$ be given by*

$$B_R^{s',s}=\left\{f\in H^{s'}(\mathbb{R}^2):\|f\|_{H^s}\le R\right\}.$$

Then, there exists $0<\varepsilon\ll1$ such that the map $\mathcal{G}_\delta^{-1}:B_R^{s',s}\to B^{s',s}_{(1+\varepsilon)R}$ is continuous with a continuous inverse $\mathcal{G}_\delta:B^{s',s}_{(1-\varepsilon)R}\to B_R^{s',s}$.

(ii) *The map $\mathcal{G}_\delta^{-1}$ is unbounded on $H^s(\mathbb{R}^2)$ for $s<-1$.*

Proof. We start with (i). It suffices to check that the quadratic term in (4.2) is a (small) bilinear map over B_R. Supposing, without loss in generality, that $|\xi| \le |\xi_1| \sim |\xi_2|$,

$$\sup_{\|g\|_{L^2}\le 1} \int_{\xi=\xi_1+\xi_2} \langle \xi\rangle^s K \mathbb{1}_{A_\delta^c}(\xi_1,\xi_2)\widehat{f_1}(\xi_1)\widehat{f_2}(\xi_2)g(\xi)d\xi_1 d\xi$$
$$\lesssim \sup_{\xi_1}\left(\int_{\xi=\xi_1+\xi_2} \mathbb{1}_{A_\delta^c}\frac{\langle\xi\rangle^{2s}}{\langle\xi_1\rangle^{2s+2}\langle\xi_2\rangle^{2s+2}}d\xi\right)^{\frac12}\|f_1\|_{H^s}\|f_2\|_{H^s} \lesssim \delta^{2+2s}\|f_1\|_{H^s}\|f_2\|_{H^s},$$

and the claim follows from (4.1).

To prove (ii), take $\widehat{f}(\xi) = \langle\xi\rangle^{-1-s-}$. Then, for $|\xi|<1$,

$$\int_{\xi=\xi_1+\xi_2} K\mathbb{1}_{A_\delta^c}(\xi_1,\xi_2)\widehat{f}(\xi_1)\widehat{f}(\xi_2)d\xi_1 \sim \int \frac{1}{\langle\xi_1\rangle^{4+2s+}}d\xi_1 = \infty,$$

since $s<-1$. □

Remark 4.4. At the endpoint $s=-1$ the gauge transform is still a bounded operator on $H^{-1}(\mathbb{R}^2)$: writing the relevant bilinear form in dual variables, the estimate reduces, via Sobolev embedding $W^{1,1}(\mathbb{R}^2)\hookrightarrow L^2(\mathbb{R}^2)$, to the standard duality $|\int\langle\partial_x\rangle^{-1}h\,g\,dx| \lesssim \|h\|_{L^1}\|g\|_{L^2}$ applied with $h=(\mathbf{P}_{>1}f)^2$.

To conclude the proof of Theorem 1.1, we proceed via a density argument as in the KdV case [11], which hinges on the following equivalence between the original and gauged flows at high regularity.

Proposition 4.5. *Let $s'\ge s>-1$ with $s'\gg 1$ sufficiently large, $R>0$, $0<\delta\ll R^{-\frac{1}{2s+2}}$, and u_0 satisfying*

$$u_0\in H^{s'}(\mathbb{R}^2) \quad \textit{and} \quad \|u_0\|_{H^s}<R. \tag{4.3}$$

Let $u\in X^{s',b}(0,T)$ be the solution to (NV) *with initial data u_0 constructed in* [3, 23, 24], *where $T>0$ is its maximal time of existence, and let $z\in X^{s',b}(0,T_{\delta,R})$ be the solution to* ($\mathcal{G}_\delta$-NV) *with initial data $z_0:=\mathcal{G}_\delta[u_0]$ provided by Proposition 4.1. Then $T\ge T_{\delta,R}$ and*

$$z(t)=\mathcal{G}_\delta[u(t)] \quad \textit{in} \quad C([0,T_{\delta,R}];H^{s'}(\mathbb{R}^2)). \tag{4.4}$$

The proof of Proposition 4.5 and the deduction of Theorem 1.1 from it occupy the remainder of this section.

The proof of Theorem 1.1 is closed by a frequency truncation/density procedure following [11, Section 5]; the NV-specific deviations are listed in the header. The parameters $s'\ge s>-1$ (with s' eventually large), $R>0$ and $0<\delta\ll R^{-1/(2s+2)}$ are fixed throughout, so that (4.1) holds.

4.1. **Truncated dynamics.** Given $n\in\mathbb{N}$, write $\mathbf{P}_n$ for the Fourier projector with symbol $\mathbb{1}_{[0,n]}(|\xi|)$, $\xi\in\mathbb{R}^2$, and consider the *frequency-truncated Novikov-Veselov equation*

$$\partial_t u_n + (\partial^3+\overline{\partial}^3)u_n + \tfrac34\mathbf{P}_n\Big(\partial(u_n\overline{\partial}^{-1}\partial u_n)+\overline{\partial}(u_n\partial^{-1}\overline{\partial}u_n)\Big)+E(\overline{\partial}^{-1}\partial^2u_n+\partial^{-1}\overline{\partial}^2u_n)=0, \tag{NV$_n$}$$

with $u_n(0)=u_0\in H^{s'}(\mathbb{R}^2)$. Since $\mathbf{P}_n$ commutes with the linear group, the resonance set A and the phase $\Psi=\Psi_0+E\Psi_E$ from Section 2.2 are insensitive to the cut-off; consequently the normal-form reduction of that section carries over to (NV$_n$) unchanged. Define

$$K_{\bullet,n} := K_\bullet\,\mathbb{1}_{[0,n]}(|\xi_\bullet|), \qquad \bullet\in\mathcal{T}^0, \tag{4.5}$$

the truncated analogue of $K_\bullet$ in (2.5), and let $\mathcal{G}_{\delta,n}$ be the *truncated gauge transform* determined by

$$\mathcal{G}_{\delta,n}^{-1}[f] := f+\mathcal{F}_z^{-1}\left(\int_{\xi=\xi_1+\xi_2} K\,\mathbb{1}_{[0,n]}(|\xi|)\,\mathbb{1}_{A_\delta^c}(\xi_1,\xi_2)\,\widehat{f}(\xi_1)\widehat{f}(\xi_2)\,d\xi_1\right),$$

i.e. (4.2) with the indicator inserted. The conclusions of Lemma 4.3 hold for $\mathcal{G}_{\delta,n}$ with δ, ε, R all uniform in n. The corresponding *truncated gauged equation* is

$$\partial_t z_n + (\partial^3 + \overline{\partial}^3) z_n + E(\overline{\partial}^{-1}\partial^2 + \partial^{-1}\overline{\partial}^2) z_n = \sum_{j=1}^{\infty} \sum_{\mathcal{T}\in \mathrm{T}_j} \sum_{\ell=1}^{5} \mathcal{N}_{\ell,n}(\mathcal{T}; z_n), \qquad (\mathcal{G}_\delta\text{-NV}_n)$$

where $\mathcal{N}_{\ell,n}(\mathcal{T})$ arises from $\mathcal{N}_\ell(\mathcal{T})$ in ($\mathcal{G}_\delta$-NV) by substituting each $K_\bullet$ in (2.25) with $K_{\bullet,n}$.

4.2. **Multilinear estimates with n-loss.** Since $\mathbb{1}_{[0,n]}$ only shrinks the multipliers, Proposition 3.1 transfers verbatim to $\mathcal{N}_{\ell,n}$. The equivalence argument also requires a coarser estimate trading the gap $s' - s$ for a polynomial factor in n — the planar version of [11, Lemma 5.2].

Lemma 4.6. *There exists $N_0 = N_0(s) \in \mathbb{N}$ (with $N_0 = 5$ admissible for $-1 < s \le 0$) such that, for $s' \ge s+2$, $j, n \in \mathbb{N}$, $\mathcal{T} \in \mathrm{T}_j$, $\ell \in \{1,\dots,5\}$, $1 \le k \le j+1$ and $f_1,\dots,f_{j+1} \in H^{s'}(\mathbb{R}^2)$,*

$$\|\mathcal{N}_{\ell,n}(\mathcal{T})[f_1,\dots,f_{j+1}]\|_{H^s(\mathbb{R}^2)} \le n^{N_0}\, C^j \,\|f_k\|_{H^s} \max_{\substack{1\le k'\le j+1\\ k'\ne k}} \left(\|f_{k'}\|_{H^{s'}} \prod_{k''\ne k,k'} \|f_{k''}\|_{H^s} \right), \quad (4.6)$$

for some $C = C(s,\delta) > 0$ independent of n, j and $\mathcal{T}$.

Proof. We import the strategy of [11, Lemma 5.2] and flag the two NV-specific entries. Combining (2.26), (2.25) and (4.5) produces the pointwise bound

$$|\operatorname{mul}_{\ell,n}(\mathcal{T})(\vec{\xi})| \le C^j\, n \max_{\bullet\in\mathcal{T}^\infty} |\xi_\bullet|^2\, \mathbb{1}_{[0,n]}(|\xi|) \prod_{\bullet\in\mathcal{T}^\infty} \frac{1}{\langle \xi_{\bullet 1}\rangle \langle \xi_{\bullet 2}\rangle}, \qquad (4.7)$$

in which $|\xi_\bullet| \le n$ for $\bullet \in \mathcal{T}^0$ has already been used; the two derivatives sitting on the largest final frequency are swallowed by $s' \ge s+2$, exactly as in [11].

The first NV-specific entry concerns the Type III multiplier $\operatorname{mul}_3(\mathcal{T})$ of (2.25): it carries an extra $E\Psi_E(\star)$ with $|\Psi_E(\star)| \lesssim |\xi_\star|$ (Section 3). Since the truncation forces $|\xi_\star| \le n$, this contributes one additional power of n beyond (4.7).

The second is dimensional: the outer integration in (4.6) takes place on $\mathbb{R}^2$. Running the $p = 2$ argument of [11, Lemma 5.2] verbatim — Cauchy-Schwarz, then successive integration along the final subtrees, then the elementary bound $\int_{|\xi|\le n} d\xi \lesssim n^d$ with $d = 2$ in place of $d = 1$ — yields (4.6) with the total loss $n^{3+d} = n^5$ (three powers from (4.7) together with the recursion, one from the Type III correction, plus $d = 2$ from the planar outer integral). The range $s \ge 0$ is reduced to the previous one via the standard inequality $\langle \xi\rangle^s \le j \max_{\bullet\in\mathcal{T}^\infty} \langle \xi_\bullet\rangle^s$ of [11, Lemma 5.2]. $\square$

4.3. **High-regularity well-posedness for the truncated gauged equation.** Following [11, Lemma 5.4], we upgrade Proposition 4.1 to a uniform-in-n statement and sharpen uniqueness at the top regularity.

Lemma 4.7. *Let $s' \ge s > -1$ with $s' \ge s+2$, $R > 0$, and $0 < \delta \ll R^{-1/(2s+2)}$. For every $n \in \mathbb{N}$ and every $z_{0,n} \in H^{s'}(\mathbb{R}^2)$ with $\|z_{0,n}\|_{H^s} \le R$, there exists a unique solution $z_n \in X^{s',b}(0, T_{\delta,R})$ to ($\mathcal{G}_\delta$-NV$_n$) with $z_n(0) = z_{0,n}$, where $T_{\delta,R}$ is the time from Proposition 4.1 (independent of n). Moreover, for $t \in [0, T_{\delta,R}]$,*

$$\|z_n(t)\|_{H^s} \le 2R, \qquad \|z_n(t)\|_{H^{s'}} \le 2\|z_{0,n}\|_{H^{s'}}, \qquad (4.8)$$

and z_n is unique in $\{z \in C([0,T_{\delta,R}]; H^{s'}(\mathbb{R}^2)) : \|z\|_{L^\infty_t H^s_x} \le 10R\}$.

Proof. The existence portion and the bounds (4.8) are a direct consequence of the proof of Proposition 4.1 together with Remark 4.2, since Proposition 3.1 controls $\mathcal{N}_{\ell,n}$ with n-independent constants.

For the sharper uniqueness statement, pick two candidates z_n, z_n' as in the hypothesis, fix $0 < T' \le T_{\delta,R}$, and combine the Duhamel formula with (4.6) to get

$$\|z_n - z_n'\|_{L^\infty_{T'} H^s_x} \lesssim T' \, n^{N_0} \left(\|z_n\|_{L^\infty H^{s'}} + \|z_n'\|_{L^\infty H^{s'}} \right) \|z_n - z_n'\|_{L^\infty_{T'} H^s_x} \sum_{j \ge 0} (CR)^j .$$

Choosing T' small enough — the threshold depends on n and on the two $H^{s'}$ norms — absorbs the right-hand side and gives $z_n = z_n'$ on $[0, T']$. A finite chain of such intervals covers $[0, T_{\delta,R}]$. □

4.4. **Equivalence of the truncated dynamics at high regularity.** For smooth data, the next lemma matches (NV_n) and ($\mathcal{G}_\delta$-NV_n) through $\mathcal{G}_{\delta,n}$ on $[0, T_{\delta,R}]$.

Lemma 4.8. *Let $s' \gg 1$ be sufficiently large, $R > 0$, $0 < \delta \ll R^{-1/(2s+2)}$, and $u_0 \in H^{s'}(\mathbb{R}^2)$ with $\|u_0\|_{H^s} < R$. Then, for every $n \in \mathbb{N}$:*

(i) *there is a unique solution $u_n \in X^{s',b}(0, T_{\delta,R})$ to* (NV_n) *on the* uniform *interval $[0, T_{\delta,R}]$, satisfying*
$$\|u_n\|_{L^\infty([0,T_{\delta,R}];H^s)} < 3R, \qquad \|u_n\|_{L^\infty([0,T_{\delta,R}];H^{s'})} < 3\|u_0\|_{H^{s'}}; \tag{4.9}$$

(ii) *if z_n is the solution to* ($\mathcal{G}_\delta$-NV_n) *furnished by Lemma 4.7 with $z_{0,n} := \mathcal{G}_{\delta,n}[u_0]$, then*
$$z_n = \mathcal{G}_{\delta,n}[u_n] \quad \text{in} \quad C([0, T_{\delta,R}]; H^{s'}(\mathbb{R}^2)); \tag{4.10}$$

(iii) $\|u_n\|_{X^{s',b}(0,T_{\delta,R})} \le C(\|u_0\|_{H^{s'}})$, *with C independent of n.*

Proof. The proof retains the three-step skeleton of [11, Lemma 5.5]; we record only the points where the NV features enter.

Step 1 (forward direction). Local well-posedness at high regularity [3, 23, 24] produces u_n on a maximal $X^{s',b}_{\mathrm{loc}}$-interval $[0, T_n)$. Put $T_{0,n} := \sup\{0 < T \le \min(T_{\delta,R}, T_n) : \|u_n\|_{L^\infty_T H^s_x} < 3R\}$ and let $v_n(t) := \mathcal{F}_z(U(-t)u_n(t))$. The algebra of Section 2.2 sees only A and $\Psi = \Psi_0 + E\Psi_E$, both preserved by $\mathbf{P}_n$, so v_n solves the truncated normal-form equation, and the four prerequisites (a)-(d) of [30, Section 4] are in place: $\mathcal{F}_z v_n(\xi)$ is C^1 in t and has rapid decay in ξ (because $s' \gg 1$), while Lemma 4.6, applied with n fixed, supplies the summability and convergence of N_n, B_n and M_n. The new Type III term $\mathcal{N}_{3,n}$ inherits the same bound; the factor $|\Psi_E(\star)| \lesssim |\xi_\star|$ has been incorporated into (2.25) once and for all. Setting $z_n' := \mathcal{G}_{\delta,n}[u_n]$, Lemma 4.3 (now read for $\mathcal{G}_{\delta,n}$) yields
$$z_n' \in C([0, T_{0,n}]; H^{s'}(\mathbb{R}^2)), \qquad \|z_n'\|_{L^\infty_{T_{0,n}} H^s_x} < (1+\varepsilon)\, 3R, \tag{4.11}$$
and the chain of term-by-term cancellations from Lemmas 2.8-2.10, identical to [11], certifies that z_n' is an integral solution to ($\mathcal{G}_\delta$-NV_n).

Step 2 (matching). Lemma 4.3 bounds $\|z_{0,n}\|_{H^s} = \|\mathcal{G}_{\delta,n}[u_0]\|_{H^s} \le R/(1-\varepsilon)$, so the z_n produced by Lemma 4.7 obeys $\|z_n\|_{L^\infty_{T_{\delta,R}} H^s} \le 2R/(1-\varepsilon)$. Coupled with (4.11), the $C_t H^{s'}$ uniqueness clause of Lemma 4.7 forces $z_n = z_n'$ in $C([0, T_{0,n}]; H^{s'})$. Reading Lemma 4.3 backwards,
$$\|u_n\|_{L^\infty_{T_{0,n}} H^s_x} \le (1+\varepsilon)\, \|z_n'\|_{L^\infty_{T_{0,n}} H^s_x} \le \tfrac{2(1+\varepsilon)}{1-\varepsilon} R < 3R,$$
provided ε is chosen small. Continuity together with the blow-up alternative of [3, 23, 24] upgrades this to $T_n \ge T_{\delta,R}$ and $T_{0,n} = T_{\delta,R}$; the second half of (4.9) is its $H^{s'}$ twin.

Step 3 ($X^{s',b}$-bound). On any sub-interval of length depending only on $M := 2\|u_0\|_{H^{s'}}$, the high-regularity well-posedness of [3, 23, 24] gives $\|u_n\|_{X^{s',b}} \le C(M)$. A deterministic concatenation, identical to [11, Lemma 5.5, Step 3] and powered by (4.9), glues these pieces uniformly in n and delivers (iii). $\square$

4.5. **Proof of Proposition 4.5.** Push $n \to \infty$ in Lemma 4.8 along the lines of [11, Proof of Proposition 5.6].

Proof of Proposition 4.5. Take u_0 as in (4.3). Denote by $z \in X^{s',b}(0, T_{\delta,R})$ the ($\mathcal{G}_\delta$-NV) solution with data $z_0 := \mathcal{G}_\delta[u_0]$ (Proposition 4.1), and let $u \in X^{s',b}(0,T)$ be the maximal (NV) solution issued from u_0 [3, 23, 24]. Lemma 4.8 furnishes u_n, z_n with $u_n(0) = u_0$ and $z_n(0) = \mathcal{G}_{\delta,n}[u_0]$. The proof reduces to the pair of convergences

$$z_n \to z \quad \text{in} \quad C([0, T_{\delta,R}]; H^{s'-1}(\mathbb{R}^2)), \tag{4.12}$$

$$\mathcal{G}_{\delta,n}[u_n] \to \mathcal{G}_\delta[u] \quad \text{in} \quad C([0, T \wedge T_{\delta,R}); H^{s'-1}(\mathbb{R}^2)). \tag{4.13}$$

Granted (4.12)-(4.13), the identity (4.10) yields $\mathcal{G}_\delta[u(t)] = z(t)$ in $H^{s'-1}$ on $[0, T \wedge T_{\delta,R})$. The $L^\infty_t H^{s'}$ control on z provided by Proposition 4.1 and Remark 4.2, transferred through Lemma 4.3 as $\|u(t)\|_{H^{s'}} \le \|z\|_{L^\infty_t H^{s'}}/(1-\varepsilon)$, prevents u from blowing up before $T_{\delta,R}$. Thus $T \ge T_{\delta,R}$ and the identity persists to all of $[0, T_{\delta,R}]$, which is (4.4).

Proof of (4.12). Subtracting the Duhamel formulations of ($\mathcal{G}_\delta$-NV) and ($\mathcal{G}_\delta$-NV$_n$) splits $z(t) - z_n(t)$ into three pieces:

$$\begin{aligned} z(t) - z_n(t) = U(t)(\mathcal{G}_\delta - \mathcal{G}_{\delta,n})[u_0] &+ \sum_{j,\mathcal{T},\ell} \int_0^t U(t-t')\big[\mathcal{N}_\ell(\mathcal{T}; z) - \mathcal{N}_{\ell,n}(\mathcal{T}; z)\big]\, dt' \\ &+ \sum_{j,\mathcal{T},\ell} \int_0^t U(t-t')\big[\mathcal{N}_{\ell,n}(\mathcal{T}; z) - \mathcal{N}_{\ell,n}(\mathcal{T}; z_n)\big]\, dt'. \end{aligned}$$

Proposition 3.1 (applied to the truncated multipliers, which it covers without change) together with the smallness of z and z_n in $L^\infty_t H^s$ bounds the third sum by $\frac{1}{2}\|z - z_n\|_{X^{s'-1,b}(0,T_{\delta,R})}$, which is absorbed. For the data term, the two presentations $u_0 = \mathcal{G}_\delta[u_0] + \mathrm{D}[\mathcal{G}_\delta[u_0], \mathcal{G}_\delta[u_0]] = \mathcal{G}_{\delta,n}[u_0] + \mathbf{P}_n \mathrm{D}[\mathcal{G}_{\delta,n}[u_0], \mathcal{G}_{\delta,n}[u_0]]$ of u_0 (D as in (4.2)), combined with the small-Lipschitz estimate on D inside the proof of Lemma 4.3, give

$$\|(\mathcal{G}_\delta - \mathcal{G}_{\delta,n})[u_0]\|_{H^{s'-1}} \lesssim \|\mathbf{P}_{>n}\mathrm{D}[\mathcal{G}_\delta[u_0], \mathcal{G}_\delta[u_0]]\|_{H^{s'-1}} \lesssim n^{-1}\|\mathrm{D}[\mathcal{G}_\delta[u_0], \mathcal{G}_\delta[u_0]]\|_{H^{s'}} \to 0.$$

Finally, $\mathcal{N}_\ell(\mathcal{T}) - \mathcal{N}_{\ell,n}(\mathcal{T})$ decomposes into finitely many operators of the same multilinear form as $\mathcal{N}_{\ell,n}$, each picking up an indicator $\mathbb{1}_{|\xi_\bullet|>n}$ on some node; Proposition 3.1 disposes of each, and trading the extra $\langle \xi_\bullet \rangle$ against the $H^{s'-1} \hookrightarrow H^{s'}$ upgrade (which z provides uniformly) creates a saving $\langle \xi_\bullet \rangle^{-1} \le n^{-1}$. The mechanism is the one of [11]; the only NV twist is that the indicator now lives on $\mathbb{R}^2$. Summing the three contributions establishes (4.12).

Proof of (4.13). The intermediate step is

$$u_n \to u \quad \text{in} \quad C([0, T \wedge T_{\delta,R}); H^{s'-1}(\mathbb{R}^2)). \tag{4.14}$$

Working on sub-intervals of length T_* controlled only by $M := 2\|u_0\|_{H^{s'}}$, the high-regularity bilinear estimates [3, 23, 24] furnish

$$\|u_n - u\|_{X^{s'-1,b}(0,T_*)} \le \tfrac{1}{2}\|u_n - u\|_{X^{s'-1,b}(0,T_*)} + \frac{C}{n} T_*^\theta \|u\|^2_{X^{s',b}(0,T_*)},$$

the factor n^{-1} originating from upgrading $\mathbf{P}_{>n}$ from $s'-1$ to s'. Chaining these sub-intervals yields (4.14) and, together with (4.9), the auxiliary bound $\|u\|_{L^\infty_t H^s} < 3R$ on $[0, T \wedge T_{\delta,R})$.

The identity (4.13) follows by inserting the two representations

$$u_n = \mathcal{G}_{\delta,n}[u_n] + \mathbf{P}_n \mathrm{D}[\mathcal{G}_{\delta,n}[u_n], \mathcal{G}_{\delta,n}[u_n]], \qquad u = \mathcal{G}_\delta[u] + \mathrm{D}[\mathcal{G}_\delta[u], \mathcal{G}_\delta[u]],$$

into

$$\mathcal{G}_{\delta,n}[u_n]-\mathcal{G}_\delta[u]=(u_n-u)-\mathbf{P}_n\big(\mathrm{D}[\mathcal{G}_{\delta,n}[u_n],\mathcal{G}_{\delta,n}[u_n]]-\mathrm{D}[\mathcal{G}_\delta[u],\mathcal{G}_\delta[u]]\big)-\mathbf{P}_{>n}\mathrm{D}[\mathcal{G}_\delta[u],\mathcal{G}_\delta[u]],$$

and treating the three summands by (4.14), by the small-Lipschitz estimate on D coming from the proof of Lemma 4.3, and by the $H^{s'}$-control of $\mathrm{D}[\mathcal{G}_\delta[u],\mathcal{G}_\delta[u]]$, respectively. □

4.6. **Conclusion of Theorem 1.1.** Proposition 4.5 states that, restricted to the ball $B_R := \{u_0 \in H^s(\mathbb{R}^2) : \|u_0\|_{H^s} < R\}$ and the uniform window $[0, T_{\delta,R}]$, the smooth (NV) flow ($u_0 \in H^{s'}(\mathbb{R}^2) \cap B_R$, $s' \gg 1$) is the conjugate, by the analytic map $\mathcal{G}_\delta$ from Lemma 4.3, of the ($\mathcal{G}_\delta$-NV) flow. The latter admits a continuous — indeed analytic — extension from $H^{s'} \cap B_R$ to the full ball $B_R \subset H^s(\mathbb{R}^2)$, via Proposition 4.1 together with the multilinear estimates of Proposition 3.1. Continuity of $\mathcal{G}_\delta$ and $\mathcal{G}_\delta^{-1}$ on $B_R \subset H^s$ (Lemma 4.3) makes the composition $\mathcal{G}_\delta^{-1} \circ \Phi^{\mathcal{G}_\delta\text{-NV}} \circ \mathcal{G}_\delta$ a continuous prolongation of the (NV) flow from $\mathcal{S}(\mathbb{R}^2) \cap B_R$ to B_R, with the existence time $T_{\delta,R}$ uniform across the ball. Density of $\mathcal{S}(\mathbb{R}^2)$ in $H^s(\mathbb{R}^2)$ identifies this extension with the limit of the smooth solutions $u_n \in C([0, T_{\delta,R}]; H^{s'})$ of [3, 23, 24]; analyticity is inherited from the factors. Theorem 1.1 is proved. □

Appendix A. Improved local well-posedness for $s > -\frac{3}{4}$

In this short appendix, we extend the well-posedness result in $H^s, s > -\frac{3}{4}$, of Adams and Grünrock for the zero-energy (NV) to the nonzero case. In particular, we are able to improve the properties of the (NV) flow constructed in Theorem 1.1.

Theorem A.1. *Fix $E \in \mathbb{R}$ and $s > -\frac{3}{4}$. Given $u_0 \in H^s(\mathbb{R}^2)$, there exists $T > 0$ and a unique $u \in X^{s,b}_T$ strong integral solution to* (NV) *with initial data u_0. Moreover, the data-to-solution map is analytic.*

As before, using the scaling invariance, it suffices to prove Theorem A.1 for $|E| < 1$. We define the bilinear operator associated with the (NV) nonlinearity (see (1.2)):

$$B[u_1, u_2]^\wedge(\xi) := \int_{\xi=\xi_1+\xi_2} K\Phi_0 u_1(\xi_1) u_2(\xi_2) d\xi_1$$

Then, using the standard theory of Fourier restriction norm spaces ([7, 8], see also the monographs [31, 42]), the proof of Theorem A.1 follows from

Proposition A.2. *Fix $E \in \mathbb{R}$ and $s > -\frac{3}{4}$. Then*

$$\|B[u_1, u_2]\|_{X^{s,-\frac{1}{2}+}} \lesssim \|u_1\|_{X^{s,\frac{1}{2}+}} \|u_2\|_{X^{s,\frac{1}{2}+}}. \tag{A.1}$$

Proof. First, we decompose B as

$$\begin{aligned} B[u_1, u_2]^\wedge(\xi) = &\int_{\xi=\xi_1+\xi_2} K\Phi \mathbb{1}_{A^c} u_1(\xi_1) u_2(\xi_2) d\xi_1 \\ &+ \int_{\xi=\xi_1+\xi_2} K(\Phi_0 \mathbb{1}_A - E\Phi_E \mathbb{1}_{A^c}) u_1(\xi_1) u_2(\xi_2) d\xi_1. \end{aligned}$$

The bilinear estimate for the last term has already been shown in Proposition 3.1 (it corresponds to the Type I term). For the remaining term, we use Lemma 3.3-(i) (together with Lemma 3.5) and prove the necessary frequency-restricted estimates.

To simplify the exposition, we take $-\frac{3}{4} < s < -\frac{1}{2}$ and consider only the worst-case scenario $|\xi_1| \sim |\xi_2| \gtrsim |\xi|$. For $\alpha \in \mathbb{R}$ and $M > 1$ with $|\alpha| \lesssim M$, we bound

$$\begin{aligned} \frac{\langle \xi \rangle^{2s} |K\Phi|^2}{\langle \xi_1 \rangle^{2s+0-} \langle \xi_2 \rangle^{2s}} \mathbb{1}_{A^c} \mathbb{1}_{|\Phi-\alpha|<M} &\lesssim |\alpha| \frac{\langle \xi \rangle^{2s+1}}{\langle \xi_1 \rangle^{2s+1-} \langle \xi_2 \rangle^{2s+1}} \mathbb{1}_{A^c} \mathbb{1}_{|\Phi-\alpha|<M} \\ &\lesssim |\alpha| \frac{1}{\langle \xi_1 \rangle^{2s+1-} \langle \xi_2 \rangle^{2s+1}} \mathbb{1}_{A^c} \mathbb{1}_{|\Phi-\alpha|<M}. \end{aligned}$$

For ξ_2 fixed, if $\xi_1 \not\simeq \pm\xi$, then, by Lemma 3.6-(i),

$$|\nabla_{\xi_1}\Phi| \gtrsim |\xi_1|^2$$

and

$$\int \frac{1}{\langle\xi_1\rangle^{2s+1-}\langle\xi_2\rangle^{2s+1}}\mathbb{1}_{|\Phi-\alpha|<M}d\xi_1 \lesssim \int \frac{1}{\langle\zeta_1\rangle^{4s+4-}}\mathbb{1}_{|\Phi-\alpha|<M}d\Phi d\zeta_1 \lesssim M,$$

for $s > -\frac{3}{4}$. On the other hand, if $\xi_1 \simeq \pm\xi$, then all frequencies are comparable and, by Lemma 3.6-(ii), there exists a change of coordinates $\xi_1 \mapsto q_1$, with jacobian $\sim |\xi_2|^2$, such that

$$\Phi = |\xi_2|^3 \operatorname{Re} q_1^2 - \phi(\xi_2).$$

Hence

$$\int_{|q_1|<1} \langle\xi_2\rangle^{-4s+0+}\mathbb{1}_{||\xi_2|^3 \operatorname{Re} q_1^2-\phi(\xi_2)-\alpha|<M}dq_1 \lesssim \langle\xi_2\rangle^{-4s-3+0+}M \lesssim M,$$

for $s > -\frac{3}{4}$. For ξ_1 (or ξ) fixed, the proof follows the same lines *mutatis mutandis*. Thus

$$\sup_{k\in\{\emptyset,1,2\}}\sup_{\xi_k}\int_{\xi=\xi_1+\xi_2} \frac{|K\Phi|^2\langle\xi\rangle^{2s}}{\langle\xi_1\rangle^{2s+0-}\langle\xi_2\rangle^{2s}}\mathbb{1}_{A^c}\mathbb{1}_{|\Phi(\bigwedge)-\alpha|<M}\, d\xi_k \lesssim \langle\alpha\rangle\, M,$$

and (A.1) follows from Lemma 3.3-(i) and Lemma 3.5. □

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